

General Constructions of Permutation-Fair Dice

Tomáš Gavenčiak
Centre for Theoretical Study
Charles University, Prague
gavenciak@cts.cuni.cz

Bernhard Haeupler
ETH Zurich
bernhard.haeupler@inf.ethz.ch

Václav Rozhoň
ETH Zurich
rozhoňv@ethz.ch

October 1, 2026

Abstract

Suppose that each of n players rolls a different die and the players are ordered by their rolls. Can the dice use pairwise disjoint labels, so that ties are impossible, while all $n!$ orders remain equally likely? We give two general constructions. An elementary symmetrization gives equal-sided dice with $(n!)^{n-1} = 2^{O(n^2 \log n)}$ faces each. A Hahn-polynomial quadrature construction, using estimates of Wilson and Zaremba, gives n equal-sided dice with $2^{O(n \log n)}$ faces each. Conversely, for every fixed $0 < \varepsilon < 1$ and all sufficiently large n , at least $(1 - \varepsilon)n$ dice in any permutation-fair collection have $2^{\Omega(\varepsilon n)}$ faces. The minimum possible size of one prescribed die is $\Theta(n^2)$, and every integer $K \geq Cn^2$ can be prescribed as its number of faces, for an absolute constant C . We also determine the exact optimum for full-set go-first fairness, construct equal-sided full-set place-fair dice with $2n^{n-2}$ faces each for $n \geq 2$, and give dice with $O(n^{13/10}(\log n)^{1/5})$ faces each whose order distribution is pointwise close to uniform, for any fixed positive relative error.

1 Introduction

There are n players, and player i is assigned a die with $k_i \geq 1$ faces. Across all dice, the faces receive the distinct labels $1, 2, \dots, \sum_i k_i$, each label appearing exactly once. The players roll independently and are ordered by their labels. We call the dice *permutation-fair* if the resulting order is uniform on the $n!$ permutations.

The problem was raised by James Ernest and has been studied extensively for small numbers of players; see, for example, [1, 2, 3, 4]. The basic existence question is deceptively simple: do permutation-fair dice exist for every n ? We answer it constructively and quantify the number of faces needed.

The dice problem can be viewed as a rigid finite-precision version of the random-key method for sampling a permutation. In the usual method one draws n independent continuous keys and sorts them; ties occur with probability zero. With finite keys, one normally tolerates a small collision probability or resamples when a collision occurs. Neither option is allowed here: every player makes one independent uniform draw, the supports are pairwise disjoint, and the induced order must be exactly uniform. This rigidity has a real cost. Our lower bound shows that, for every fixed $0 < \varepsilon < 1$, all but an ε -fraction of the players need supports of size $2^{\Omega(\varepsilon n)}$, and hence $\Omega(\varepsilon n)$ bits of entropy per draw. This is a lower bound for this one-shot, disjoint-support representation, not for arbitrary algorithms that sample random permutations.

1.1 Our results

A three-die example. It is useful to begin with a concrete instance. Consider the string

$$s = \boxed{2} \boxed{1} \boxed{3} \boxed{1} \boxed{2} \boxed{2} \boxed{2} \boxed{2} \boxed{1} \boxed{3} \boxed{1} \boxed{2}.$$

The j -th box specifies which die owns the face labelled j ; red, green, and blue represent dice 1, 2, 3, respectively. Thus the string represents

$$\boxed{1} : \{2, 4, 9, 11\}, \quad \boxed{2} : \{1, 5, 6, 7, 8, 12\}, \quad \boxed{3} : \{3, 10\}.$$

A joint roll chooses one occurrence of each box color, and their left-to-right order is the order of the rolled labels. Each of the six possible orders occurs exactly eight times as a subsequence of s . Since there are $4 \cdot 6 \cdot 2 = 48$ possible rolls, every order has probability $8/48 = 1/6$.

More generally, read all face labels in increasing order and record the owner of each face. This turns any collection of dice into a string s , in which symbol i occurs k_i times. If $C_\pi(s)$ denotes the number of occurrences of a permutation π as a subsequence of s , then

$$\Pr[\text{order} = \pi] = \frac{C_\pi(s)}{\prod_{i=1}^n k_i}.$$

Consequently, the dice are permutation-fair exactly when all $n!$ permutations occur equally often in s . This correspondence replaces a probabilistic question about dice by a combinatorial question about strings.

The elementary construction. Our first construction starts from $12 \cdots n$ and repeatedly concatenates all $n!$ relabellings of the string obtained so far. For $n = 3$, the first round is

$$u_1 = \boxed{1} \boxed{2} \boxed{3},$$

$$u_2 = \boxed{1} \boxed{2} \boxed{3} \mid \boxed{1} \boxed{3} \boxed{2} \mid \boxed{2} \boxed{1} \boxed{3} \mid \boxed{2} \boxed{3} \boxed{1} \mid \boxed{3} \boxed{1} \boxed{2} \mid \boxed{3} \boxed{2} \boxed{1}.$$

The string u_2 has equal counts for all two-symbol patterns. Fix any ordering $S_3 = \{\tau_1, \dots, \tau_6\}$. Concatenating the six relabellings in that order gives

$$u_3 = u_2^{\tau_1} \circ u_2^{\tau_2} \circ \cdots \circ u_2^{\tau_6},$$

which is fully permutation-fair and has 36 occurrences of each symbol. In general, one round enforces symmetry for patterns one symbol longer, so $n - 1$ rounds suffice.

Theorem 1.1 (Elementary construction (informal version of Theorem 2.11)). *For every n , there are n equal-sided permutation-fair dice with $2^{O(n^2 \log n)}$ faces each.*

This argument is elementary and robust, but it pays for symmetry by using all $n!$ relabellings at every step. Our second construction avoids this loss.

The Hahn-polynomial construction. Here is again the three-symbol picture first. Using Simpson's degree-three quadrature rule, the successive strings are

$$v_1 = \boxed{1} \boxed{1} \boxed{1}, \quad v_2 = \boxed{2} \mid \boxed{1} \boxed{1} \boxed{1} \mid \boxed{2} \boxed{2} \boxed{2} \boxed{2} \mid \boxed{1} \boxed{1} \boxed{1} \mid \boxed{2},$$

$$v_3 = \boxed{3} \boxed{3} \mid v_2 \mid \boxed{3} \boxed{3} \boxed{3} \boxed{3} \boxed{3} \boxed{3} \boxed{3} \mid v_2 \mid \boxed{3} \boxed{3}.$$

The first insertion uses gap lengths $(1, 4, 1)$, and the second uses $(2, 8, 2)$. The final string has twelve occurrences of each symbol, and each full permutation occurs 288 times.

In general, we add one new symbol at a time by separating copies of the old string with runs of the new symbol. The condition on the gap lengths is exactly a quadrature identity for Bernstein polynomials. Using Hahn polynomials and estimates of Wilson and Zaremba, we construct positive rational weights on an equally spaced grid. Their common denominator has only $O(n \log n)$ bits, and the same rule can be reused at every insertion step.

Theorem 1.2 (Hahn construction (informal version of Theorem 3.6)). *For every n , there are n equal-sided permutation-fair dice with $2^{O(n \log n)}$ faces each.*

Lower bounds. Restricting a fair string to any p symbols, where p is prime, forces p to divide the product of their multiplicities. Applying this observation simultaneously for many primes gives the following result.

Theorem 1.3 (Lower bound (informal version of Theorem 4.1)). *Every permutation-fair collection of n dice contains a die with $2^{\Omega(n)}$ faces.*

Thus, if $f(n)$ is the smallest possible value of the largest die in a permutation-fair collection, then

$$2^{\Omega(n)} \leq f(n) \leq 2^{O(n \log n)}.$$

The precise lower bound is stronger: for every fixed $0 < \varepsilon < 1$, it forces $2^{\Omega(\varepsilon n)}$ faces on at least $(1 - \varepsilon)n$ dice.

The smallest individual die. The exponential bound applies to most dice, but a prescribed die can be much smaller. Let $g(n)$ be the minimum possible number of faces of one die in a permutation-fair family of n dice.

Theorem 1.4 (Optimal individual size (informal version of Theorem 4.3)). *We have $g(n) = \Theta(n^2)$.*

The construction controls the size of one die; the minimum total size remains open.

Weaker and approximate notions. Sections 5 and 6 contain three further results.

- (i) **Go-first fairness (informal version of Theorem 5.1).** Among full-set go-first-fair strings, allowing different multiplicities, the minimum possible total length is exactly $n(n + 1)/2$.
- (ii) **Place fairness (informal version of Theorem 5.4).** For every $n \geq 2$, there are n equal-sided, full-set place-fair dice with $2n^{n-2}$ faces each.
- (iii) **Approximate permutation fairness (informal version of Theorem 6.1).** For every $n \geq 3$ and $0 < \varepsilon \leq 1$, there are equal-sided dice with

$$O\left(n^{13/10}(\log n)^{1/5}\varepsilon^{-2/5}\right)$$

faces each that give every permutation probability within relative error ε of $1/n!$.

Roadmap. Section 2 develops the string representation and the elementary symmetrization construction. Section 3 gives the sharper Hahn-polynomial construction. Section 4 proves the lower bounds and determines the minimum size of one prescribed die. Section 5 treats go-first and place fairness, and Section 6 treats approximate permutation fairness. We conclude with open problems in Section 7. The technical estimates for the approximate construction are proved in Appendix A.

1.2 Related work

Permutation-fair, place-fair, and go-first-fair dice have been investigated for small n in [1, 2, 3, 4]. The insertion framework underlying our second construction originates in this literature; an early online account attributes an inductive version to Paul Vanetti [5]. Ford et al. give five equal 60-sided hereditarily place-fair dice, unequal-sided permutation-fair examples, and five equal 180-sided permutation-fair dice [3]. Purcell develops the symmetrization construction further and uses it to construct five equal 120-sided permutation-fair dice [4]. A computational construction due to Paul Meyer gives five equal 60-sided permutation-fair dice [6]. We are not aware of a previous treatment of the exact full-set, unequal-multiplicity go-first optimum proved in Section 5.

A weaker string problem asks only that every permutation occur at least once as a subsequence. This has a long history [7, 8, 9, 10, 11, 12, 13, 14]; see Engen and Vatter [15] for a survey. A superficially similar but different problem asks for an ordinary permutation in which all order-isomorphic patterns of a fixed length occur equally often. Beniamini, Lavee, and Linial settle that problem, including nonexistence from pattern length four onward [16]; their host permutation has no repeated letters, so that obstruction does not apply here.

There is also an established literature on *fair words*. In Černý’s terminology [17, 18], a word is fair if $C_{ab} = C_{ba}$ for every pair of distinct letters; the binary case goes back at least to Prodinger [19]. Among equal-composition words this is exactly our $(2, n)$ -fairness. Černý also studied reversal symmetry of longer scattered subwords [20]; see Salomaa [21] for related subword-balance questions and Richomme [22] for a recent account. To avoid confusing this pairwise notion with equality of all full-permutation counts, we use *permutation-fair string* for the latter in prose while retaining the compact notation n -fair and (k, n) -fair.

The same pairwise condition appears in work on tied and balanced nontransitive dice, often through the same word encoding [23, 24, 25, 26, 27]. Those papers prescribe pairwise win probabilities, rather than the joint distribution of the full order, and are therefore adjacent to but weaker than the permutation-fair problem.

Our second construction also draws on classical equidistant quadrature and the Hahn polynomials. Wilson gave the discrete least-squares projection formula for the quadrature weights and studied equidistant quadrature [28, 29, 30]; his Hahn estimates control the continuous integrals of the basis polynomials [31]. Huybrechs revisited Wilson’s stable high-order rules on equidistant points [32], and later least-squares quadrature work treats scattered nodes and other weight functions [33, 34]. Classical positive quadrature goes back at least to Fejér and Shohat [35, 36]. For positivity we use Zaremba’s sharper bound at the integer nodes [37]; Dette later obtained bounds on the whole interval [38]. Our use of these ingredients is arithmetic: we give an explicit common-denominator estimate and show how one rule can be reused in every insertion step.

More broadly, existence theorems for unweighted designs on connected spaces go back to Seymour and Zaslavsky [39], with modern quantitative results due to Kane [40]. Chebyshev-type interval quadrature for the uniform measure has quadratic-order real-node bounds

going back to Bernstein, with later sharp formulations by Kuijlaars and by Gilboa and Peled [41, 42, 43, 44]; see also Peled’s general-measure bounds [45]. Imposing rational nodes leads to the rational-design problem of Cui, Xia, and Xiang [46], built in part on Hilbert–Kamke methods [47], a related arithmetic variant of the quadrature problem. Our construction uses real quadrature nodes and corrects their rational approximation by changing the distributions of the other dice.

Approximate dice connect to classical shuffling laws. The periodic word $(12 \cdots n)^m$ realizes an inverse riffle shuffle, whose probabilities are described by descent enumeration [48, 49]. Repeating the palindrome $12 \cdots n n \cdots 21$ realizes a shelf shuffle; Diaconis, Fulman, and Holmes [50] analyze its maximal relative error and establish the $n^{3/2}$ scale for fixed accuracy. We independently relabel the palindrome blocks and bound the resulting error using multilinear hypercontractivity [51, 52] and a multinomial collision estimate.

2 Permutation-fair strings and elementary construction

2.1 Fair strings and their properties

Write $[n] = \{1, \dots, n\}$. A *pattern* is a string with no repeated letter. Let $\Pi_{n,k}$ be the set of all length- k patterns over $[n]$, and let $\Pi_n = \Pi_{n,n}$. For a pattern π and a string s , write $C_\pi(s)$ for the number of occurrences of π as a subsequence of s . We put $C_\emptyset(s) = 1$.

Definition 2.1 (String representation). Let $D_1, \dots, D_n$ be dice whose face labels partition $[K]$. Their *string representation* is the string $s \in [n]^K$ defined by $s[a] = i$ when label a is a face of D_i .

Definition 2.2 (Permutation-fair string). A string s over $[n]$ in which every letter occurs at least once is *n-fair*, or *permutation-fair*, if $C_\pi(s)$ has the same value for every $\pi \in \Pi_n$.

Lemma 2.3. *A set of dice is permutation-fair if and only if its string representation is permutation-fair.*

Proof. Choose one face from each die. Sorting the chosen labels produces an occurrence of a unique pattern $\pi \in \Pi_n$ in the string representation. Conversely, every such occurrence selects exactly one face from every die. Since all $\prod_i k_i$ face selections have the same probability, the orders are equiprobable exactly when their subsequence counts are equal. $\square$

Relabelling the alphabet preserves fairness. Restricting to fewer letters does as well: after fixing one face of every deleted die, summing over the positions of those faces shows that all orders of the retained letters have the same count. We record this together with another useful operation.

Claim 2.4. *Let s be n-fair.*

- (i) *Relabelling the letters of s by a permutation of $[n]$ preserves fairness, and restricting s to any subset of its letters produces a permutation-fair string on that subset.*
- (ii) *Replacing every occurrence of a fixed letter i by a run of q copies of i , where $q \in \mathbb{N}$, preserves fairness.*

Proof. Relabelling merely permutes the pattern counts. The restriction statement follows by the preceding summation argument. For (ii), every occurrence of a full pattern in s has exactly q lifts to the enlarged string, one for each copy of its unique occurrence of i . $\square$

In particular, arbitrary permutation-fair dice may be made equal-sided, although this generic operation is much more expensive than the equal-sided construction of Section 3.

Corollary 2.5. *If s is n -fair, there is an n -fair string s' in which every letter has the same multiplicity and*

$$|s'| \leq |s|^n.$$

Proof. If letter i occurs c_i times, blow it up by $\prod_{j \neq i} c_j$. Every letter then occurs $\prod_j c_j$ times, so

$$|s'| = n \prod_j c_j \leq n \left(\frac{|s|}{n} \right)^n \leq |s|^n.$$

□

Full fairness determines all partial-pattern counts.

Lemma 2.6 (Partial-pattern count). *Let s be n -fair, let C be the common count of a full pattern, and let $c_i = C_i(s)$. If $\pi \in \Pi_{n,k}$ uses the set $S \subseteq [n]$, then*

$$C_\pi(s) = \frac{(n!/k!)C}{\prod_{\ell \notin S} c_\ell}. \quad (1)$$

Proof. Count pairs consisting of an occurrence of π and one chosen occurrence of each omitted letter. Directly, their number is $C_\pi(s) \prod_{\ell \notin S} c_\ell$. Sorting all chosen positions gives a full pattern whose restriction to S is π . There are $n!/k!$ such full patterns, each with count C , proving (1). □

Corollary 2.7. *If s is n -fair, then $s \circ s$, and hence every positive power s^m , is n -fair.*

Proof. For $\pi \in \Pi_n$, split an occurrence according to the number j of letters chosen from the first copy:

$$C_\pi(s \circ s) = \sum_{j=0}^n C_{\pi[1:j]}(s) C_{\pi[j+1:n]}(s).$$

For fixed j , applying Lemma 2.6 to the two complementary patterns makes the product independent of π : its denominator contains each c_i exactly once. The result follows, and iteration gives s^m . □

For the first construction, it is convenient to ask for fairness at every pattern length.

Definition 2.8. A string s over $[n]$ is (k, n) -fair if $C_\pi(s)$ is the same for all $\pi \in \Pi_{n,k}$. It is $(\leq k, n)$ -fair if it is (j, n) -fair for every $0 \leq j \leq k$.

The concatenation of two $(\leq k, n)$ -fair strings is again $(\leq k, n)$ -fair: split an occurrence at the concatenation point and use fairness of the two pieces at the corresponding lengths. Equal-composition permutation-fair strings automatically satisfy this stronger property.

Lemma 2.9. *If s is n -fair and every letter occurs equally often, then s is $(\leq n, n)$ -fair.*

Proof. In (1), the denominator depends only on the number of omitted letters, not on their identities. Thus every pattern count depends only on its length. □

2.2 The first construction: symmetrization

We first give an elementary construction. Rather than adding letters one at a time, it progressively enforces symmetry for longer patterns.

If $\tau \in \Pi_n$ is viewed as a permutation of $[n]$, let s^τ denote the string obtained from s by replacing every letter i by $\tau(i)$.

Lemma 2.10 (Symmetrization; cf. [4]). *Let $k < n$. From any $(\leq k, n)$ -fair string s of length ℓ , one can construct a $(\leq k+1, n)$ -fair string of length $\ell n!$.*

Proof. List the $n!$ permutations as $\tau_1, \dots, \tau_{n!}$, in any order, and set

$$s' = s^{\tau_1} \circ s^{\tau_2} \circ \dots \circ s^{\tau_{n!}}.$$

Fix $t \leq k+1$, a pattern $\pi \in \Pi_{n,t}$, and an occupancy vector $(q_1, \dots, q_{n!})$ specifying how many consecutive letters of π are selected from each block. If every $q_a \leq k$, the contribution of this occupancy vector is a product of counts of patterns of lengths q_a in relabellings of s . Since s is $(\leq k, n)$ -fair, that product depends only on the vector, not on π .

The only remaining case is $t = k+1$ and all $k+1$ selected letters lie in one block. Summed over these $n!$ possible blocks, the contribution is

$$\sum_{\tau \in \Pi_n} C_\pi(s^\tau).$$

As τ ranges over all permutations, $\tau^{-1}(\pi)$ ranges uniformly over the length- $(k+1)$ patterns: each such pattern occurs $(n-k-1)!$ times. This sum is therefore independent of π . Summing over all occupancy vectors proves that s' is $(\leq k+1, n)$ -fair. $\square$

Theorem 2.11 (Elementary symmetrization). *For every n , there are n equal-sided permutation-fair dice with $(n!)^{n-1}$ faces each. Equivalently, there is a $(\leq n, n)$ -fair string of length*

$$n(n!)^{n-1} = 2^{O(n^2 \log n)}.$$

Proof. The string $s_1 = 12 \dots n$ is $(\leq 1, n)$ -fair. Apply Lemma 2.10 $n-1$ times. Each application multiplies the length by $n!$. $\square$

This proves the qualitative existence of permutation-fair dice for every n . The next construction improves the exponent and keeps every die the same size throughout.

3 The second construction: Hahn quadrature

We now add one letter at a time. Fix the final target n , and let s be an equal-composition r -fair string, where $r < n$. Given N copies of s and nonnegative integers $k_0, \dots, k_N$, form

$$T = (r+1)^{k_0} \circ s \circ (r+1)^{k_1} \circ s \circ \dots \circ s \circ (r+1)^{k_N}. \quad (2)$$

There are $N+1$ gaps, numbered $0, \dots, N$, around N copies of s .

3.1 The insertion step is a quadrature problem

Let every old symbol occur F times in s , and put $K = \sum_i k_i$ and $w_i = k_i/K$. The Bernstein polynomials [53]

$$b_{r,j}(x) = \binom{r}{j} x^j (1-x)^{r-j}, \quad 0 \leq j \leq r,$$

satisfy

$$\int_0^1 b_{r,j}(x) dx = \frac{1}{r+1}. \quad (3)$$

Lemma 3.1 (Insertion criterion). *If*

$$\sum_{i=0}^N w_i b_{r,j}(i/N) = \frac{1}{r+1} \quad (0 \leq j \leq r), \quad (4)$$

then the string T in (2) is $(r+1)$ -fair.

Proof. By Lemma 2.9, s is $(\leq r, r)$ -fair; so is each power s^a . Consequently, for any fixed ordered set of j old symbols,

$$C_\pi(s^i) = \frac{(iF)^j}{j!},$$

because all $j!$ orders partition the $(iF)^j$ choices of one occurrence of each symbol. The analogous suffix count is $((N-i)F)^{r-j}/(r-j)!$.

Fix a full pattern in which the new symbol has rank $j+1$. If its chosen occurrence of $r+1$ lies in gap i , the number of choices for the old symbols is therefore

$$\frac{(iF)^j}{j!} \frac{((N-i)F)^{r-j}}{(r-j)!}.$$

After summing over all k_i occurrences in every gap, its total count is

$$\begin{aligned} \sum_{i=0}^N k_i \frac{(iF)^j ((N-i)F)^{r-j}}{j!(r-j)!} &= \frac{KF^r N^r}{r!} \sum_{i=0}^N w_i b_{r,j}(i/N) \\ &= \frac{KF^r N^r}{(r+1)!}, \end{aligned}$$

which is independent of both j and the order of the old symbols. $\square$

Thus it suffices to find positive rational weights w_i that integrate all polynomials of degree at most n on $[0, 1]$. The same rule will then satisfy (4) at every stage $r < n$.

Remark 3.2 (Why positivity needs care). A Riemann sum is close to the integral, but correcting its moments at only $n+1$ selected grid points does not automatically preserve nonnegativity. For example, in degree 2, correcting the uniform weights $1/m$ at $0, 1/3, 2/3$ gives at $1/3$ the weight

$$-\frac{1}{2m} + \frac{3}{2m^2},$$

which is negative for every $m > 3$. The Hahn-basis projection below spreads the correction across the entire grid and gives a uniform positivity estimate.

3.2 Hahn polynomials on an equally spaced grid

Set

$$N = n(n+2), \quad N+1 = (n+1)^2. \quad (5)$$

For $0 \leq k \leq n$, define the discrete Chebyshev (Hahn) polynomial

$$Q_k(x) = Q_k(x; 0, 0, N+1) = {}_3F_2\left(\begin{matrix} -k, -x, k+1 \\ 1, -N \end{matrix}; 1\right). \quad (6)$$

In Wilson's normalization [31], $Q_k(0) = 1$, and

$$Q_k(x) = \sum_{a=0}^k (-1)^a \binom{k}{a} \binom{k+a}{a} \frac{x^a}{N^a}. \quad (7)$$

The polynomials $Q_0, \dots, Q_n$ are orthogonal for the uniform measure on $\{0, 1, \dots, N\}$. Define the unnormalized squared norm and the continuous integral

$$H_k = \sum_{i=0}^N Q_k(i)^2, \quad I_k = \int_0^N Q_k(x) dx.$$

The standard Hahn norm formula (see the primary formulas in [54, 38], as well as [55, Sec. 9.5, Eq. (9.5.2)] and [56, Table 18.19.1], translated from their largest-grid-index convention to Wilson's number-of-grid-points convention) specializes to

$$H_k = \frac{N+1}{2k+1} \frac{(N+2)(N+3) \cdots (N+k+1)}{N(N-1) \cdots (N-k+1)} = \frac{(N+k+1)!(N-k)!}{(2k+1)(N!)^2}. \quad (8)$$

In particular,

$$H_k \geq \frac{N+1}{2k+1}. \quad (9)$$

Lemma 3.3 (Zaremba's nodal bound). *For every $0 \leq k \leq n$ and every integer $i \in \{0, \dots, N\}$,*

$$|Q_k(i)| \leq 1. \quad (10)$$

Proof. In the largest-grid-index convention used by Zaremba and Dette, our $Q_k(x; 0, 0, N+1)$ is $Q_k(x, 0, 0, N)$. Zaremba's nodal estimate [37] (see also [38, Remark 3.3]) gives $|Q_k(i)| \leq 1$ at every integer $0 \leq i \leq N$ whenever

$$k(k+1) \leq N.$$

Here

$$k(k+1) \leq n(n+1) < n(n+2) = N.$$

The case $k = 0$ is immediate. □

Wilson's aim was to quantify how closely Hahn polynomials on a uniform grid imitate their Jacobi limits [31]. He developed an expansion in inverse powers of the grid size and used it to compare discrete and continuous least-squares expansions. In the Legendre case he isolated the integrals I_k : they measure the failure of the nonconstant discrete orthogonal basis to have zero continuous mean, and they also enter quadrature formulas on equidistant nodes. In our argument they are precisely the coefficients that must be controlled in the projection weights below.

Lemma 3.4 (Wilson's integral estimate). *Let $N \geq 1$, let $0 \leq k \leq N$, and let Q_k and I_k be as above. If k is odd, then $I_k = 0$. If $k \geq 2$ is even and $N + 1 \geq (k + 1)^2$, then*

$$0 < -I_k \leq \frac{N+1}{N} \left(1 + \frac{(k-1)(k+2)}{4(N-1)} \right). \quad (11)$$

Consequently, for the choice $N = n(n+2)$,

$$|I_k| < \frac{3}{2} \quad (1 \leq k \leq n). \quad (12)$$

Proof. The reflection identity $Q_k(N-x) = (-1)^k Q_k(x)$ gives $I_k = (-1)^k I_k$, and hence the odd case. Wilson writes our I_k as $I_k(N+1)$. For even k , his lower estimate shows that $-I_k > 0$, while his non-asymptotic upper estimate is exactly (11); see [31, Sec. 3, Eqs. (3.1) and (3.14)].

For the consequence, when $n = 1$ the only index is $k = 1$, which is odd. If $n \geq 2$, then $N + 1 = (n + 1)^2 \geq (k + 1)^2$, and

$$\frac{N+1}{N} \leq \frac{9}{8}, \quad \frac{(k-1)(k+2)}{4(N-1)} \leq \frac{(n-1)(n+2)}{4(N-1)} < \frac{1}{4}.$$

Thus $|I_k| < 45/32 < 3/2$ for even k , while the odd integrals vanish. $\square$

Wilson's proof is genuinely finite-grid rather than merely asymptotic. Using a recurrence due to Levit, he rewrites I_k as a weighted integral of a neighboring Hahn polynomial; its zeros yield convexity and endpoint bounds, and a coefficient-ratio argument turns those bounds into a geometric-series estimate. The resulting quadratic relation between grid size and degree is exactly what is needed here, because k grows with n .

3.3 Positive exact weights

The following formula is the discrete least-squares projection formula for quadrature weights used by Wilson [29]: it expands the integral functional in the orthogonal basis $Q_0, \dots, Q_n$. For $0 \leq i \leq N$, put

$$W_i = \sum_{k=0}^n \frac{I_k}{H_k} Q_k(i). \quad (13)$$

Proposition 3.5 (Exactness and positivity). *The weights $W_0, \dots, W_N$ are strictly positive and satisfy*

$$\sum_{i=0}^N W_i p(i) = \int_0^N p(x) dx \quad \text{for every polynomial } p \text{ with } \deg p \leq n. \quad (14)$$

Moreover, $\sum_i W_i = N$.

Proof. For $0 \leq \ell \leq n$, discrete orthogonality gives

$$\sum_{i=0}^N W_i Q_\ell(i) = \sum_{k=0}^n \frac{I_k}{H_k} \sum_{i=0}^N Q_k(i) Q_\ell(i) = I_\ell.$$

The Q_ℓ form a basis for the polynomials of degree at most n , which proves exactness. Taking $p = 1$ yields $\sum_i W_i = N$.

It remains to prove positivity. Since $Q_0 = 1$, $I_0 = N$, and $H_0 = N + 1$, the $k = 0$ term in (13) is $N/(N + 1)$. If $n = 1$, then $I_1 = 0$, so every $W_i = 3/4 > 0$. Suppose now

that $n \geq 2$, and put $m = \lfloor n/2 \rfloor$. Since all odd I_k vanish, Lemma 3.3, (12), and (9) give, uniformly in $0 \leq i \leq N$,

$$\begin{aligned} \left| W_i - \frac{N}{N+1} \right| &\leq \sum_{\substack{2 \leq k \leq n \\ k \text{ even}}} \frac{|I_k| |Q_k(i)|}{H_k} \\ &< \frac{3}{2(N+1)} \sum_{j=1}^m (4j+1) \\ &= \frac{3(2m^2 + 3m)}{2(N+1)} \\ &\leq \frac{3n(n+3)}{4(N+1)}. \end{aligned}$$

Consequently,

$$W_i > \frac{N - 3n(n+3)/4}{N+1} = \frac{n(n-1)}{4(N+1)} > 0.$$

□

After scaling from $[0, N]$ to $[0, 1]$, the normalized weights $w_i = W_i/N$ satisfy

$$\sum_{i=0}^N w_i q(i/N) = \int_0^1 q(x) dx \quad (\deg q \leq n). \quad (15)$$

3.4 A small common denominator

Positivity alone is not enough: the gap lengths must be integers. We now bound a common denominator explicitly. Let

$$A_k = N^{\underline{k}} = N(N-1) \cdots (N-k+1), \quad B_k = (N+2)(N+3) \cdots (N+k+1),$$

with $A_0 = B_0 = 1$, and let

$$L = \text{lcm}(1, 2, \dots, n+1).$$

Formula (7) shows that

$$A_k Q_k(x) \in \mathbb{Z}[x], \quad Q_k(i) \in A_k^{-1} \mathbb{Z} \quad (i \in \mathbb{Z}). \quad (16)$$

Integrating the integer-coefficient polynomial $A_k Q_k(x)$ term by term between integer endpoints introduces only denominators $1, \dots, k+1$. Thus

$$I_k \in (A_k L)^{-1} \mathbb{Z}. \quad (17)$$

On the other hand, (8) says

$$H_k = \frac{(N+1)B_k}{(2k+1)A_k}.$$

Combining these facts gives

$$\frac{I_k}{H_k} Q_k(i) \in ((N+1)LA_k B_k)^{-1} \mathbb{Z}. \quad (18)$$

Since $A_k \mid A_n$ and $B_k \mid B_n$, the single integer

$$M = (N+1)LA_n B_n \quad (19)$$

clears all the weights:

$$MW_i \in \mathbb{Z}_{>0} \quad (0 \leq i \leq N). \quad (20)$$

Finally, $L \leq (n+1)!$, $A_n \leq N^n$, and $B_n \leq (2N)^n$. Therefore

$$M \leq 2^{n+1} (n+1)^{n+1} N^{2n+1} = 2^{O(n \log n)}. \quad (21)$$

3.5 Reusing one rule at every induction level

Theorem 3.6 (Hahn construction). *For every $n \geq 1$, there are n equal-sided permutation-fair dice with F_n faces each, where, for $N = n(n+2)$,*

$$F_n = (N+1) \operatorname{lcm}(1, \dots, n+1) N^n \cdot (N+2)(N+3) \cdots (N+n+1) N^{n-1}. \quad (22)$$

In particular,

$$F_n = 2^{O(n \log n)}.$$

The corresponding string is $(\leq n, n)$ -fair.

Proof. Use the positive weights and the denominator M constructed above, and start with

$$s_1 = 1^M.$$

Inductively suppose s_r is r -fair and every symbol occurs

$$F_r = MN^{r-1}$$

times. Around N copies of s_r , put in gap i a run of the new symbol of length

$$k_i = F_r W_i. \quad (23)$$

These numbers are positive integers because F_r is divisible by M . By Proposition 3.5,

$$\sum_{i=0}^N k_i = F_r \sum_{i=0}^N W_i = NF_r.$$

Each old symbol also occurs NF_r times in the N copies. Thus every symbol in the new string occurs

$$F_{r+1} = NF_r = MN^r$$

times. The normalized gap weights are $k_i/(NF_r) = W_i/N$, and (15) is exact for all degree- r Bernstein polynomials. Lemma 3.1 therefore proves that the new string is $(r+1)$ -fair.

After $n-1$ insertion steps, every symbol occurs MN^{n-1} times. Substituting the definition (19) of M gives (22). Using $N+1 \leq 2N$ and the estimates above,

$$F_n \leq 2^{n+1} (n+1)^{n+1} N^{3n} = 2^{O(n \log n)}.$$

The total string length is $nF_n = 2^{O(n \log n)}$. Since all letter multiplicities are equal, Lemma 2.9 also makes the final string $(\leq n, n)$ -fair. $\square$

The rule in Theorem 3.6 uses exactly $N+1 = (n+1)^2$ grid points. The quadratic grid changes the constants in the construction but not its $2^{O(n \log n)}$ asymptotic face bound.

Remark 3.7 (A three-die check). The three-die example in Section 1.1 uses Simpson's degree-three rule on 0, 1, 2 in place of the general-purpose rule:

$$(W_0, W_1, W_2) = \left(\frac{1}{3}, \frac{4}{3}, \frac{1}{3} \right), \quad N = 2, \quad M = 3.$$

4 Sizes of permutation-fair dice

Divisibility across restrictions forces exponentially many faces on most dice. We first prove that statement, then determine the minimum possible size of one prescribed die.

Theorem 4.1 (Many dice are large). *There is an absolute constant $c > 0$ such that, for every fixed $0 < \varepsilon < 1$ and all sufficiently large n , every permutation-fair collection with face counts $k_1, \dots, k_n$ satisfies*

$$|\{i \in [n] : k_i \geq 2^{c\varepsilon n}\}| \geq (1 - \varepsilon)n.$$

4.1 Divisibility across restrictions

The first argument is arithmetic. Divisibility obstructions are classical in the fair-dice literature; in the equal-sided setting, the corresponding primordial condition is recorded by Purcell [4]. Applying the same idea after restriction to many different subsets gives a useful bound even when the dice have unequal numbers of faces.

Lemma 4.2. *Every n -fair string has length $2^{\Omega(n)}$.*

Proof. Let $c_i = C_i(s)$. Restrict s to a set S of p letters, where $p \leq n$ is prime. The restricted string remains p -fair. If C is its common full-pattern count, then

$$\prod_{i \in S} c_i = p!C.$$

Hence at least one c_i , $i \in S$, is divisible by p . Since this holds for every p -subset, at most $p - 1$ of the n numbers c_i can fail to be divisible by p . Consequently,

$$\prod_{i=1}^n c_i \geq \prod_{\substack{p \leq n/2 \\ p \text{ prime}}} p^{n/2}.$$

The primordial estimate $\prod_{p \leq x} p = 2^{\Theta(x)}$ [57] now gives $\prod_i c_i = 2^{\Omega(n^2)}$. Therefore $\max_i c_i \geq 2^{\Omega(n)}$, and the string is at least that long. $\square$

Proof of Theorem 4.1. Let $r = \lfloor \varepsilon n \rfloor$, and restrict a fair string to its r least frequent symbols. By Lemma 4.2, the restricted string has length at least $2^{c_0 r}$ for some absolute $c_0 > 0$, so one of these r symbols occurs at least $2^{c_0 r}/r$ times. Each of the other $n - r$ symbols occurs at least as often. For fixed ε and sufficiently large n ,

$$\frac{2^{c_0 r}}{r} \geq 2^{(c_0/4)\varepsilon n} \quad \text{and} \quad n - r \geq (1 - \varepsilon)n.$$

This proves Theorem 4.1 with $c = c_0/4$. $\square$

It is important not to overinterpret this argument. For every subset $S \subseteq [n]$, restriction and permutation fairness imply the basic divisibility condition

$$|S|! \mid \prod_{i \in S} c_i.$$

These conditions alone are compatible with one prescribed die having a single face: the multiplicity vector

$$(1, n!, n!, \dots, n!)$$

satisfies all of them. This is not a construction of permutation-fair dice; it only shows that divisibility alone cannot prove a nontrivial lower bound for a die specified in advance.

4.2 The optimal size of one die

The arithmetic lower bound forces almost all dice to be exponentially large, but it leaves room for a prescribed die to be much smaller. The following theorem determines the optimal order of its size.

Theorem 4.3 (The size of one die). *Every die in a permutation-fair family of $n \geq 2$ dice has at least $(n-1)^2/200$ faces. Conversely, there is an absolute constant $C > 0$ such that, for every $n \geq 2$ and every integer $K \geq Cn^2$, there is a permutation-fair family in which a specified die has exactly K faces. The remaining $n-1$ dice can all have the same finite number of faces.*

Thus, if $g(n)$ is the smallest possible number of faces of one specified die in a permutation-fair family of n dice, then

$$g(n) = \Theta(n^2).$$

The construction below gives no quantitative bound for the other dice, and hence no improvement in the minimum total size.

4.2.1 A universal quadratic lower bound

Fix a die D_0 with sorted faces $x_1 < \dots < x_K$, and put $m = n-1$. For each other die D_i , define

$$p_i(q) = \Pr(D_i < x_q), \quad 1 \leq i \leq m, \quad 1 \leq q \leq K.$$

Each $p_i(q)$ is nondecreasing in q . Independence, followed by permutation fairness restricted to D_0 and any subset of the other dice, gives

$$\frac{1}{K} \sum_{q=1}^K \prod_{i \in S} p_i(q) = \frac{1}{|S|+1} \quad (S \subseteq [m]). \quad (24)$$

Indeed, the left side is the probability that D_0 beats every die indexed by S . We will use these identities with general positive weights w_q summing to one in place of $1/K$. By linearity, their weighted version implies

$$\sum_{q=1}^K w_q F(p_1(q), \dots, p_m(q)) = \int_0^1 F(t, \dots, t) dt \quad (25)$$

for every multi-affine polynomial F , meaning one whose degree in each variable is at most one. We write $\mathbb{E}_w$ for the weighted sum on the left.

Lemma 4.4 (Paired thresholds). *Suppose the vectors $p(q) \in [0, 1]^m$ are coordinatewise nondecreasing in q . For distinct indices j, k and any $t \in (0, 1)$, there are thresholds α, β such that*

$$\{\alpha, \beta\} = \{t, b\}, \quad 0 \leq b \leq t, \quad (p_j(q) - \alpha)(p_k(q) - \beta) \geq 0 \quad \text{for every } q.$$

Proof. Adjoin the all-zero and all-one vectors and interpolate successive vectors linearly to form a continuous coordinatewise nondecreasing path. Stop at the first point where either coordinate j or coordinate k reaches t , and let α, β be their values there. One equals t , and the other is at most t . Every original row before this point lies below both thresholds, and every row after it lies above both. Thus the two differences have the same weak sign. The interpolating points are used only to choose thresholds; they carry no weight in (25). $\square$

We need the following elementary facts about quadrature for the uniform measure on $[0, 1]$. Their proofs are included to specify the endpoint estimates used below.

Lemma 4.5 (Gaussian and Radau rules). *The D -node Gaussian rule has positive weights, is exact through degree $2D - 1$, and has largest node ξ_D satisfying*

$$1 - \xi_D \leq \frac{\pi^2}{4(D+1)^2}. \quad (26)$$

The right Radau rule with interior nodes $t_1 < \dots < t_e$ and endpoint 1 has positive weights and is exact through degree $2e$. Its endpoint weight and interior nodes satisfy

$$\omega_{\text{end}} = \frac{1}{(e+1)^2}, \quad \sum_{\ell=1}^e \frac{1}{1-t_\ell} = \frac{e(e+2)}{2}. \quad (27)$$

Proof. Let P_r denote the Legendre polynomial normalized by $P_r(1) = 1$. The Gaussian nodes are the shifted roots of P_D . Polynomial division by the node polynomial and orthogonality give exactness through degree $2D - 1$; applying the rule to squared cardinal polynomials gives positive weights. The unshifted nodes are the eigenvalues of the symmetric D -by- D Jacobi matrix with zero diagonal and off-diagonal entries

$$\frac{k}{\sqrt{(2k-1)(2k+1)}} \geq \frac{1}{2}, \quad 1 \leq k < D.$$

The matrix with all these entries replaced by $1/2$ has largest eigenvalue $\cos(\pi/(D+1))$, with a positive sine eigenvector. Using that vector in the Rayleigh quotient shows that the largest unshifted Gaussian node is at least $\cos(\pi/(D+1))$. Rescaling to $[0, 1]$ and using $1 - \cos x \leq x^2/2$ proves (26).

For the Radau rule, put

$$J_e(t) = \sum_{r=0}^e (2r+1)P_r(2t-1).$$

Legendre orthogonality gives the reproducing identity $\int_0^1 J_e(t)f(t)dt = f(1)$ for $\deg f \leq e$. Consequently J_e is orthogonal to all polynomials of degree at most $e-1$ for the weight $1-t$, and has e distinct roots in $(0, 1)$. Interpolation at these roots and at 1 gives a rule exact through degree $2e$: division by $(t-1)J_e(t)$ leaves a remainder of degree at most e , and the quotient has degree at most $e-1$. The integral of the quotient term vanishes by orthogonality. Applying the rule to $(1-t)(J_e(t)/(t-t_\ell))^2$ shows that every interior weight is positive. At the endpoint, exactness on J_e^2 gives

$$\omega_{\text{end}} J_e(1)^2 = \int_0^1 J_e(t)^2 dt = J_e(1) = (e+1)^2.$$

Finally, the Legendre endpoint derivative identity gives

$$\frac{J'_e(1)}{J_e(1)} = \frac{\sum_{r=0}^e (2r+1)r(r+1)}{(e+1)^2} = \frac{e(e+2)}{2}.$$

The logarithmic derivative of J_e at 1 is $\sum_{\ell=1}^e (1-t_\ell)^{-1}$, proving the remaining identity. $\square$

Lemma 4.6 (The last row). *Suppose $w_q > 0$, $\sum_q w_q = 1$, the vectors $p(q) \in [0, 1]^m$ are coordinatewise nondecreasing, and (25) holds. Write $a_i = p_i(K)$. Then*

$$a_i \geq 1 - \frac{\pi^2}{(m+1)^2} \quad (1 \leq i \leq m), \quad w_K \leq \frac{200}{m^2}.$$

Proof. Fix i , put $d = \lfloor (m-1)/2 \rfloor$, and choose d disjoint pairs of coordinates other than i . Let $\xi_1 < \dots < \xi_D$ be the nodes of the Gaussian rule with $D = d+1$ nodes. For pair ℓ , apply Lemma 4.4 at ξ_ℓ , omitting the largest node ξ_D . Let T be the product of the resulting paired factors. It is multi-affine and nonnegative on every row, and

$$P(t) := T(t, \dots, t) = \prod_{\ell=1}^d (t - \xi_\ell)(t - b_\ell), \quad b_\ell \leq \xi_\ell < \xi_D.$$

Since $a_i - p_i(q) \geq 0$ and coordinate i is absent from T , (25) and Gaussian exactness yield

$$0 \leq \mathbb{E}_w[(a_i - p_i)T(p)] = \int_0^1 (a_i - t)P(t) dt = \omega_D(a_i - \xi_D)P(\xi_D).$$

Here $\omega_D > 0$ and $P(\xi_D) > 0$, also when $d = 0$. Therefore $a_i \geq \xi_D$. Since $D+1 \geq (m+1)/2$, we obtain

$$1 - a_i \leq \delta := \frac{\pi^2}{(m+1)^2} \quad \text{for every } i.$$

For $m \geq 10$, put $e = \lfloor m/10 \rfloor$ and choose e disjoint pairs of coordinates. Apply Lemma 4.4 at the interior nodes of the right Radau rule, one node for each pair. Again let T be the product of the paired factors, and write

$$P(t) = T(t, \dots, t) = \prod_{\ell=1}^e (t - t_\ell)(t - b_\ell), \quad b_\ell \leq t_\ell.$$

Its degree is $2e \leq m$. Since P vanishes at the interior Radau nodes, exactness gives

$$\mathbb{E}_w T(p) = \frac{P(1)}{(e+1)^2}. \quad (28)$$

Put $S = \sum_{\ell} (1 - t_\ell)^{-1} = e(e+2)/2$. The inequalities $e \leq m/10$ and $e+2 \leq 3m/10$ imply

$$\delta S \leq \frac{3\pi^2}{200} < \frac{1}{4}.$$

In particular $\delta < 1 - t_\ell$ for every ℓ , so all paired thresholds lie strictly below all a_i . Each paired factor at the last row, divided by its value at the all-one vector, is at least $(1 - \delta/(1 - t_\ell))^2$. Using $\prod_j (1 - z_j) \geq 1 - \sum_j z_j$ for $0 \leq z_j \leq 1$, we get

$$\frac{T(a)}{P(1)} \geq \prod_{\ell=1}^e \left(1 - \frac{\delta}{1 - t_\ell}\right)^2 \geq 1 - 2\delta S > \frac{1}{2}.$$

All rows contribute nonnegatively to (28). Hence $w_K \leq 2/(e+1)^2 \leq 200/m^2$. For $m < 10$, the same bound follows directly from $w_K \leq 1$. $\square$

Proof of the lower bound in Theorem 4.3. Apply Lemma 4.6 to the comparison probabilities of any chosen die, with $w_q = 1/K$ and $m = n-1$. Equation (24) supplies the required identities, and the conclusion is $K \geq (n-1)^2/200$. $\square$

4.2.2 A construction with a prescribed number of faces

The upper bound starts with a real equal-weight quadrature with K nodes. We approximate the nodes by rationals and correct the resulting order probabilities by changing the distributions of the other dice. The corrections have disjoint supports and vanishing moments, so their effect on every order probability is exactly linear. Finally, we replace the corrected continuous distributions by finite uniform dice without changing the number K of distinguished faces.

Lemma 4.7 (A regular starting quadrature). *There is an absolute constant C_0 such that, for every $m \geq 1$ and every integer $K \geq C_0(m+1)^2$, there are distinct nodes $0 < u_1 < \dots < u_K < 1$ satisfying*

$$\frac{1}{K} \sum_{q=1}^K p(u_q) = \int_0^1 p(t) dt \quad (\deg p \leq m).$$

We may take C_0 large enough that $K \geq m^2$.

Proof. Theorem 1.1 of Gilboa and Peled [44], applied to the constant weight, gives an equal-weight quadrature of degree d with exactly K nodes for every integer $K \geq C_1 d^2$. In their notation, the theorem bounds the threshold $\bar{N}_w(d)$ for all sufficiently large cardinalities, as well as the minimum cardinality $N_w(d)$. For the constant weight their local-mass quantity is of order d^2 ; affine rescaling gives the assertion on $[0, 1]$.

Apply this result at degree $2m+2$, with the prescribed K . The rule has at least $m+2$ distinct support points: otherwise the square of its support-vanishing polynomial would have degree at most $2m+2$, zero quadrature sum, and positive integral. At least m support points therefore lie in $(0, 1)$. Choose one node at each of m distinct interior points as a pivot. The Jacobian of the first m moment sums with respect to these nodes has entries su_q^{s-1}/K , $1 \leq s \leq m$, and is an invertible scaled Vandermonde matrix. The analytic implicit function theorem allows arbitrary sufficiently small perturbations of all other nodes, with compensating changes of the pivots preserving these m moments.

The pivots remain distinct and interior. Move each free endpoint node slightly into $(0, 1)$. Collisions between free nodes form proper hyperplanes in the free parameters. If a free node initially coincides with a pivot, varying that free node alone changes the matching pivot with derivative -1 , because their moment-Jacobian columns are equal. The difference of the two coordinates thus has nonzero derivative at the initial point, so their collision set is a proper analytic zero set. Initially distinct coordinates remain distinct in a sufficiently small neighborhood. A finite union of these proper zero sets cannot cover the open set of inward perturbations. Choose a collision-free perturbation and sort the nodes. All K nodes are now distinct and interior, and the moments through degree m are unchanged. $\square$

Lemma 4.8 (Local moment corrections). *For distinct interior nodes u_q , choose pairwise disjoint open intervals $I_q \subset (0, 1)$ around them. After shrinking these intervals if necessary, there are neighborhoods $V_q \subset I_q$ of u_q and bounded piecewise-constant functions $h_{q,t}$, $t \in V_q$, supported in I_q , such that*

$$\int_0^t h_{q,t}(x)g(x) dx = g(t), \quad \int_t^1 h_{q,t}(x)g(x) dx = -g(t) \quad (\deg g \leq m-1). \quad (29)$$

Their breakpoints are rational, and their values are rational when t is rational. Their coefficient vectors depend continuously on t , and the functions are uniformly bounded for t near u_q . In particular, their full moments through degree $m-1$ vanish.

Proof. Inside each I_q , choose m disjoint rational-endpoint intervals strictly to the left of u_q , and another m strictly to its right. Choose V_q between these two groups. For any m ordered disjoint intervals $J_1 < \dots < J_m$ of positive length, the matrix

$$M_{ra} = \int_{J_a} x^r dx, \quad 0 \leq r \leq m-1, \quad 1 \leq a \leq m,$$

is rational and invertible. Indeed, multilinearity of the determinant gives

$$\det M = \int_{J_1 \times \dots \times J_m} \prod_{a < b} (x_b - x_a) dx_1 \dots dx_m > 0.$$

Solve the left interval system with right-hand side $(1, t, \dots, t^{m-1})^T$, and the right interval system with its negative. Use the solutions as the values of $h_{q,t}$ on those intervals, and set it to zero elsewhere. The equations give (29) on the monomial basis. The matrices are fixed, so their solutions have all the stated rationality, continuity, and boundedness properties. $\square$

The next lemma explains why these local corrections can repair every full ordering simultaneously. Write $\mathcal{P}_r$ for the real polynomials of degree at most r .

Lemma 4.9 (The response of order probabilities). *Choose $t_q \in V_q$, pairwise disjoint sets $Q_1, \dots, Q_m \subseteq [K]$, and coefficients $\delta_{j,q}$ such that*

$$f_j(x) = 1 + \sum_{q \in Q_j} \delta_{j,q} h_{q,t_q}(x), \quad j \in [m],$$

are nonnegative densities. Let $X_0, \dots, X_m$ be independent, with X_0 uniform on $t_1, \dots, t_K$ and X_j having density f_j for $j \in [m]$. Define linear functionals on $\mathcal{P}_{m-1}$ by

$$B_j(a) = \frac{1}{K} \sum_{q \in Q_j} \delta_{j,q} a(t_q).$$

If all B_j equal the same functional B , then every specified full ordering in which X_0 has exactly k predecessors has probability

$$\frac{1}{K} \sum_{q=1}^K p_k(t_q) + B(p'_k), \quad p_k(t) = \frac{t^k(1-t)^{m-k}}{k!(m-k)!}. \quad (30)$$

Proof. Fix the full ordering and condition on $X_0 = t_\ell$. Expand the product of the densities of the other variables. Choosing the constant term from every density gives probability $p_k(t_\ell)$.

Consider a term choosing bumps for $b \geq 1$ variables, and expand their sums over the bump indices. These indices are distinct because the sets Q_j are disjoint. The chosen supports are therefore disjoint and ordered. If their order is inconsistent with the specified full ordering, the term is zero. Otherwise, integrate out the variables whose selected density contribution was the constant 1. The resulting kernel is a product of powers of gaps between successive chosen variables, t_ℓ , and the endpoints 0, 1, divided by factorials. On the relevant ordered region it is a polynomial of total degree $m - b$.

Only the support I_ℓ can contain t_ℓ . If $b \geq 2$, at least one chosen bump has its whole support on one side of t_ℓ . Because the support intervals are disjoint and ordered, varying this coordinate within its support crosses neither another chosen coordinate nor t_ℓ . Hence the same gap-polynomial formula holds throughout that support. For fixed values of the

other chosen variables, its integration is over its entire support against a polynomial of degree at most $m - b \leq m - 1$. Its vanishing full moments make the term zero. The same argument eliminates a one-bump term unless its index is ℓ . Thus only one-bump terms cut by t_ℓ can survive.

For such a bump from a variable before X_0 , let a variables precede that variable and b lie between it and X_0 , so $a + b = k - 1$. Its kernel, with $t = t_\ell$, is

$$\frac{x^a(t-x)^b(1-t)^{m-k}}{a!b!(m-k)!}.$$

The left identity in (29) evaluates this polynomial at $x = t$, giving zero unless $b = 0$. Hence only the variable immediately preceding X_0 contributes, with polynomial

$$A_k(t) = \frac{t^{k-1}(1-t)^{m-k}}{(k-1)!(m-k)!} \quad (k \geq 1).$$

For a variable after X_0 , the kernel is

$$\frac{t^k(x-t)^a(1-x)^b}{k!a!b!}, \quad a + b = m - k - 1.$$

The right identity in (29) takes its negative value at $x = t$. Only the variable immediately following X_0 contributes, with the negative of

$$C_k(t) = \frac{t^k(1-t)^{m-k-1}}{k!(m-k-1)!} \quad (k \leq m-1).$$

Define $A_0 = 0$ and $C_m = 0$. Both polynomials have degree at most $m-1$, and $A_k - C_k = p'_k$. Averaging over t_ℓ applies the functional B_j of the predecessor to A_k and that of the successor to C_k , omitting a term when the neighbor is absent. Since all these functionals equal B , the total correction is $B(p'_k)$, as claimed. $\square$

Proof of the construction in Theorem 4.3. Put $m = n - 1$ and fix the prescribed integer $K \geq Cn^2$, where C is large enough for Lemma 4.7. Choose its nodes u_q and the corresponding neighborhoods and corrections from Lemma 4.8. As $K \geq m^2$, we can choose disjoint sets $Q_1, \dots, Q_m$, each of size m .

Approximate each u_q by a rational $t_q \in V_q$, retaining their strict order. For $1 \leq s \leq m$, put

$$\mu_s(t) = \frac{1}{K} \sum_{q=1}^K t_q^s, \quad a_s(t) = \frac{1/(s+1) - \mu_s(t)}{s}.$$

For each $j \in [m]$, solve the rational Vandermonde system

$$\frac{1}{K} \sum_{q \in Q_j} \delta_{j,q} t_q^{s-1} = a_s(t) \quad (1 \leq s \leq m). \quad (31)$$

The distinct nodes make its matrix invertible. Its solution depends continuously on all the nodes near u and tends to zero as $t \rightarrow u$, since the starting quadrature gives $a_s(u) = 0$. The bumps are uniformly bounded near these starting nodes. A sufficiently close rational approximation therefore ensures that

$$f_j(x) = 1 + \sum_{q \in Q_j} \delta_{j,q} h_{q,t_q}(x) \geq \frac{1}{2} \quad \text{for every } x \in [0, 1].$$

The full zeroth moments of the bumps vanish, so each f_j integrates to one. These are positive piecewise-constant densities with rational breakpoints and rational values.

Equation (31) makes all functionals B_j in Lemma 4.9 equal to one functional B . Checking on monomials and constants shows that, for every $p \in \mathcal{P}_m$,

$$B(p') = \int_0^1 p(t) dt - \frac{1}{K} \sum_{q=1}^K p(t_q).$$

The lemma therefore gives every full ordering probability

$$\int_0^1 \frac{t^k (1-t)^{m-k}}{k!(m-k)!} dt = \frac{1}{(m+1)!}.$$

We have obtained a permutation-fair continuous model with one uniform K -point variable.

It remains to realize this model by finite uniform dice. Partition $[0, 1]$ at all breakpoints of the f_j and at every t_q . Ignore the boundary points, which have probability zero for the continuous variables. On each remaining interval J_g , all the densities are positive rational constants. Thus the masses

$$\alpha_{j,g} = \int_{J_g} f_j(x) dx$$

are positive rationals with $\sum_g \alpha_{j,g} = 1$ for every j . Choose a common denominator D for these masses. By Theorem 2.11, take an equal-sided permutation-fair family on the m other dice, with F faces each. For each interval J_g , make a copy of its fair word and replace each occurrence of letter j by $D\alpha_{j,g}$ consecutive copies. Claim 2.4(ii) shows that the resulting block is permutation-fair; it has $FD\alpha_{j,g}$ faces of die j . Concatenate these blocks in interval order and insert one distinguished face at each boundary t_q . Every other die now has exactly FD faces, whereas the distinguished die still has exactly K .

For each other die, the probabilities of selecting the different intervals are exactly $\alpha_{j,g}$, as in the continuous model. Conditional on these interval choices, variables in different intervals have a determined relative order. Within a common interval, the continuous variables are independent uniforms on that interval, so their relative order is uniform. The corresponding discrete variables also have a uniform relative order, by fairness of the block and its restrictions, using Claim 2.4(i). Different interval groups involve disjoint sets of independent dice, so their internal orders combine independently. Consequently the full order distribution, including the distinguished faces at the interval boundaries, is unchanged. Assigning consecutive distinct integer labels to the concatenated word produces the required finite dice. $\square$

5 Weaker notions of exact fairness

For this section, an *outcome* means one uniformly chosen occurrence of every letter. A string is *go-first fair* if each letter is equally likely to be the leftmost selected occurrence. It is *place fair* if, for every rank, each letter is equally likely to occupy that rank. These are full-set notions: only the outcome involving all n dice is constrained. We call either property *hereditary* if it is required after restriction to every nonempty subset of the dice. In contrast, permutation fairness is automatically hereditary by Claim 2.4(i). The full-set/hereditary distinction matters for both constructions and lower bounds below.

5.1 The exact full-set go-first optimum

Theorem 5.1. *Among full-set go-first-fair strings on n letters, allowing unequal multiplicities, the minimum length is*

$$L_{\text{GF}}(n) = \frac{n(n+1)}{2}.$$

Proof. For the upper bound, take

$$s_n = 12 \cdots n(n-1)(n-2)^2 \cdots 2^{n-2} 1^{n-1}. \quad (32)$$

Thus letter i occurs $m_i = n - i + 1$ times: once in the initial block and $n - i$ times in the descending staircase suffix. For $i < n$, letter i wins precisely when it selects its initial occurrence and every letter $j < i$ selects one of its later occurrences. Hence

$$\Pr(i \text{ wins}) = \frac{1}{n - i + 1} \prod_{j=1}^{i-1} \frac{n - j}{n - j + 1} = \frac{1}{n}.$$

Letter n , which has a single occurrence, wins with probability

$$\prod_{j=1}^{n-1} \frac{n - j}{n - j + 1} = \frac{1}{n}.$$

The string is therefore go-first fair and has length $n(n+1)/2$.

For the matching lower bound, let an arbitrary go-first-fair string have letter multiplicities m_i . Order its letters as $i_1, \dots, i_n$ according to the positions $p_1 < \dots < p_n$ of their first occurrences. Let E_k be the event that every earlier letter i_j , $j < k$, selects an occurrence after p_k . If E_k fails, the winning selected occurrence belongs to one of $i_1, \dots, i_{k-1}$. Fairness therefore gives

$$\Pr(E_k^c) \leq \frac{k-1}{n}, \quad \Pr(E_k) \geq \frac{n-k+1}{n}.$$

If i_k selects its first occurrence and E_k occurs, then i_k wins. These two events are independent, and consequently

$$\frac{1}{n} = \Pr(i_k \text{ wins}) \geq \frac{\Pr(E_k)}{m_{i_k}} \geq \frac{n-k+1}{nm_{i_k}}.$$

Thus $m_{i_k} \geq n - k + 1$. Summing over k proves the lower bound. $\square$

The hereditary versions of go-first and place fairness are arithmetically more expensive even when the dice may have different numbers of faces. Let die i have m_i faces. For every prime $p \leq n$ and every p -element set S , hereditary go-first fairness forces $p \mid \prod_{i \in S} m_i$. Hence at most $p-1$ face counts can fail to be divisible by p , and therefore

$$\prod_{i=1}^n m_i \geq \prod_{\substack{p \leq n \\ p \text{ prime}}} p^{n-p+1} = 2^{\Theta(n^2)}, \quad \max_i m_i \geq 2^{\Omega(n)}. \quad (33)$$

The same argument at any fixed rank applies to hereditary place fairness. When all m_i equal m , the argument recovers $\text{rad}(n!) \mid m$.

5.2 An exact place-fair construction

The Hahn construction already gives place-fair dice, as a consequence of full permutation fairness, with $2^{O(n \log n)}$ faces per die. Thus the construction below does not improve the asymptotic scale merely by weakening the notion of fairness. Its interest is instead structural: the rank counts turn out to be controlled by equalities of moments through degree $n - 2$, bringing the classical problem of equal sums of like powers directly into the design of dice. It gives a simple explicit construction independent of the Hahn machinery, and every improvement in consecutive multigrade partitions automatically yields place-fair dice with fewer faces.

A useful route to exact place fairness is therefore a consecutive Prouhet–Tarry–Escott partition. We first isolate the dice-specific reduction, then retain Prouhet’s especially transparent digit-sum construction [58], and finally invoke the stronger partition theorem of Choudhry [59].

Prouhet’s construction and its relation to the later Tarry–Escott problem are discussed by Lehmer and Wright [60, 61, 62]. Roberts studied partitions of consecutive integers and their digit structure [63, 64], while Bolker et al. connect the problem to generalized Thue–Morse sequences [65]. For two classes, optimizing the interval length is also the problem of constructing Littlewood polynomials with a high-order zero at one; see [66, 67, 68, 69]. These results motivate shorter colorings, but the reduction to place-fair dice below works for any number of colors.

Lemma 5.2 (Prouhet digit-sum identity [58]). *Let $n \geq 2$, $m = n^{n-1}$, and write $t = \sum_{\ell=0}^{n-2} d_\ell n^\ell$, where $0 \leq d_\ell < n$. Put*

$$c(t) = \sum_{\ell=0}^{n-2} d_\ell \pmod{n}.$$

For every $0 \leq q \leq n - 2$, the sum

$$\sum_{\substack{0 \leq t < m \\ c(t)=a}} t^q$$

is independent of $a \in \mathbb{Z}/n\mathbb{Z}$.

Proof. Let ω be a primitive n -th root of unity. For $1 \leq h < n$,

$$\sum_{t=0}^{m-1} \omega^{hc(t)} e^{tx} = \prod_{\ell=0}^{n-2} \sum_{d=0}^{n-1} (\omega^h e^{n^\ell x})^d.$$

Every factor on the right has a zero at $x = 0$, since $\omega^h \neq 1$. Thus the product has a zero of order at least $n - 1$. Taking derivatives of orders $q = 0, 1, \dots, n - 2$ at zero shows

$$\sum_{t=0}^{m-1} \omega^{hc(t)} t^q = 0 \quad (0 \leq q \leq n - 2).$$

Fourier inversion on $\mathbb{Z}/n\mathbb{Z}$ now says exactly that the power sums of all color classes are equal. $\square$

Theorem 5.3 (Moment-coloring reduction). *Let $n \geq 2$ and $m \geq 1$ be integers, and suppose that there is a map*

$$c : \{0, 1, \dots, m - 1\} \longrightarrow \mathbb{Z}/n\mathbb{Z}$$

such that, for every $0 \leq q \leq n - 2$, the sum

$$\sum_{\substack{0 \leq t < m \\ c(t)=a}} t^q$$

is independent of $a \in \mathbb{Z}/n\mathbb{Z}$. Then there are n equal-sided, full-set place-fair dice with m faces each.

Proof. Fix a coloring c satisfying the hypothesis. For $0 \leq t < m$, let B_t be the cyclic block of length n whose letter at within-block position $r \in [n]$ is the unique $i \in [n]$ satisfying

$$i \equiv r - c(t) \pmod{n}.$$

Concatenate the blocks $B_0 B_1 \cdots B_{m-1}$. Every block contains every letter once, so every die has m faces.

Fix a face of letter i in block t , and suppose that its within-block position is r . For each of the $r - 1$ letters before it in that block, there are $t + 1$ faces before the chosen face and $m - t - 1$ after it. For each of the remaining $n - r$ letters, the corresponding numbers are t and $m - t$. Therefore the generating polynomial for the number of other selected faces lying before this face is

$$F_{t,r}(z) = ((m - t - 1) + (t + 1)z)^{r-1} ((m - t) + tz)^{n-r}. \quad (34)$$

For $0 \leq k < n$, put $P_{r,k}(t) = [z^k]F_{t,r}(z)$. This is a polynomial in t of degree at most $n - 1$. Its leading coefficient is

$$[z^k](z - 1)^{n-1},$$

which is independent of r .

Letter i occupies position r in block t exactly when $c(t) \equiv r - i \pmod{n}$. Hence its number of outcomes at rank $k + 1$ is

$$C_{i,k} = \sum_{r=1}^n \sum_{\substack{0 \leq t < m \\ c(t) \equiv r-i \pmod{n}}} P_{r,k}(t). \quad (35)$$

For powers t^q with $q \leq n - 2$, the hypothesis makes the inner class sum independent of its residue. The coefficient of t^{n-1} is independent of r , while the classes $r - i$, $r \in [n]$, run through all residues. Thus (35) is independent of i . For each outcome exactly one die occupies rank $k + 1$, so every die occupies it in exactly m^n/n outcomes. This holds for every rank, proving place fairness. $\square$

The digit-sum coloring of Lemma 5.2, followed by Theorem 5.3, gives the especially transparent bound $m = n^{n-1}$. The following stronger Prouhet–Tarry–Escott construction improves this by a factor of $n/2$.

Theorem 5.4 (Improved place-fair upper bound). *For every $n \geq 2$, there are n equal-sided, full-set place-fair dice with*

$$m = 2n^{n-2}$$

faces each. For $n = 1$, one face suffices.

Proof. For $n = 2$, the digit-sum coloring of Lemma 5.2 has $m = 2$. Now let $n \geq 3$. Choudhry [59, Theorem 6] partitions the consecutive integers

$$1, 2, \dots, 2n^{n-2}$$

into n equally sized classes whose sums of q -th powers agree for every $1 \leq q \leq n - 2$. Equal class sizes give the assertion for $q = 0$, and translating every integer by -1 preserves equality of all moments through degree $n - 2$. The translated classes therefore satisfy the hypothesis of Theorem 5.3 with $m = 2n^{n-2}$. The same consecutive-interval partition theorem was independently obtained by Pollack and Treviño [70]. $\square$

6 Approximate permutation fairness

Exact permutation fairness forces exponentially many faces on most dice. We now ask how small the dice can be if every order is allowed a relative error of at most ε . Thus the required inequality is

$$|n!P(\pi) - 1| \leq \varepsilon \quad \text{for every } \pi \in S_n. \quad (36)$$

This controls even individually rare orders. In particular, it implies $\|P - U\|_1 \leq \varepsilon$ and $d_{\text{TV}}(P, U) \leq \varepsilon/2$, where U is uniform on S_n .

6.1 Classical shuffle constructions

The periodic word $(12 \cdots n)^m$ realizes an inverse m -riffle shuffle: each letter independently chooses a uniform level in $[m]$, and ties are resolved in increasing letter order. If $d = \text{des}(\pi)$, the classical P -partition count gives

$$P(\pi) = \frac{1}{m^n} \binom{m+n-1-d}{n}, \quad n!P(\pi) = \prod_{k=-d}^{n-1-d} \left(1 + \frac{k}{m}\right) \quad (m \geq n). \quad (37)$$

See Stanley [48] and Bayer and Diaconis [49]. The product decreases with d . Its two extreme values show that $m \geq n(n-1)/\varepsilon$ suffices for (36) when $0 < \varepsilon \leq 1$: the increasing-order ratio is at most $e^{\varepsilon/2} \leq 1 + \varepsilon$, and the decreasing-order ratio is at least $1 - \varepsilon/2$. Conversely, for $0 < \varepsilon < 1$ the decreasing order forces $m \geq n$ and

$$m \geq \frac{n(n-1)}{2[-\log(1-\varepsilon)]}.$$

Thus this periodic family requires $\Theta(n^2/\varepsilon)$ faces per die for $0 < \varepsilon \leq 1/2$.

A better classical construction repeats a palindrome:

$$(12 \cdots n n \cdots 21)^m.$$

Its order distribution is exactly the shelf shuffle of Diaconis, Fulman, and Holmes [50], as given by their Description 1. Each die independently chooses one of the $2m$ monotone runs uniformly; reading the chosen letters groups them by run, in increasing order within odd runs and decreasing order within even runs. This is precisely their packet construction, so their Theorem 3.1 applies with the valleys of the output permutation itself. Their Theorem 3.4 gives, for fixed $c > 0$ and $m \sim cn^{3/2}$,

$$\max_{\pi \in S_n} |n!P(\pi) - 1| \rightarrow e^{1/(12c^2)} - 1.$$

Consequently, for each fixed positive error, $O(n^{3/2})$ faces per die suffice. Our construction improves this bound by independently relabelling the palindrome blocks.

6.2 Independent palindrome blocks

For a permutation ρ , let ρ^{rev} be its reversal. Choose $\rho_1, \dots, \rho_m$ independently and uniformly from S_n , and form the word

$$W = (\rho_1 \rho_1^{\text{rev}}) \circ \dots \circ (\rho_m \rho_m^{\text{rev}}). \quad (38)$$

Every resulting die has exactly $2m$ faces. The randomness here chooses the dice; after the word has been chosen, their order distribution P_W is fixed.

Theorem 6.1 (Approximate fairness from random palindromes). *There is an absolute constant C such that, for every $n \geq 3$ and $0 < \varepsilon \leq 1$, if*

$$m \geq C n^{13/10} (\log n)^{1/5} \varepsilon^{-2/5},$$

then the random word (38) satisfies (36) simultaneously for all $\pi \in S_n$ with probability at least $1/2$. In particular, there exist n equal-sided dice with

$$O\left(n^{13/10} (\log n)^{1/5} \varepsilon^{-2/5}\right)$$

faces each satisfying (36).

The palindrome structure makes every pair of letters exactly fair within each block. Thus a block contributes no error unless at least three dice choose faces in it. Independent relabelling centers the remaining errors; the proof controls their joint contribution, including interactions between several occupied blocks.

Here is the exact expansion behind that argument. Fix a target order σ . Let $L = (L_1, \dots, L_m)$ have the multinomial distribution with n trials and m equally likely cells, and split σ into consecutive segments A_i of lengths L_i . For an ordered list A of distinct letters, write $q_i(A)$ for the probability that these letters appear in the prescribed order when each chooses one of its two occurrences in block i . Put

$$\delta_i(A) = |A|! q_i(A) - 1,$$

with $\delta_i(\emptyset) = 0$. Summing over the chosen block indices gives the identity

$$n! P_W(\sigma) = \mathbb{E}_L \prod_{i=1}^m (1 + \delta_i(A_i)). \quad (39)$$

For $|A| \leq 2$, $\delta_i(A) = 0$ identically. For every fixed A , uniform relabelling gives $\mathbb{E}_{\rho_i} \delta_i(A) = 0$.

The following estimate controls the whole expansion. Its proof, given in Appendix A, combines hypercontractivity with a bound on multinomial collision probabilities. The L^p norm is over the independent random permutations defining W .

Lemma 6.2 (Uniform moment estimate). *Let $n \geq 3$, $2 \leq p \leq n^2$, and $m \geq 8n^{11/10} p^{1/5}$. For every $\sigma \in S_n$,*

$$\|n! P_W(\sigma) - 1\|_p \leq 32 \frac{n^{11/4} \sqrt{p}}{m^{5/2}}.$$

Proof of Theorem 6.1. Set

$$p = \max \left\{ 2, \frac{\log(2n!)}{\log 3} \right\}.$$

For $n \geq 3$, we have $p \leq n^2$ and $p = O(n \log n)$. Choosing the absolute constant C sufficiently large ensures $m \geq 8n^{11/10}p^{1/5}\varepsilon^{-2/5}$. Lemma 6.2 then gives, for every σ ,

$$\|n!P_W(\sigma) - 1\|_p \leq \frac{32}{8^{5/2}}\varepsilon < \frac{\varepsilon}{3}.$$

Markov's inequality and a union bound imply

$$\Pr(\exists \sigma \in S_n : |n!P_W(\sigma) - 1| > \varepsilon) \leq n!3^{-p} \leq \frac{1}{2}.$$

Thus a single sampled collection satisfies all the required inequalities with the stated probability. $\square$

6.3 A lower bound for total size

For fixed error, quadratic total length is necessary even if only the first-place marginals are required to be approximately fair. Suppose each die is first with probability at most $(1 + \varepsilon)/n$. Order the letters as $i_1, \dots, i_n$ according to the positions $p_1 < \dots < p_n$ of their first occurrences. Let E_k be the event that every earlier letter i_j , $j < k$, chooses an occurrence after p_k . If E_k fails, the first selected occurrence belongs to one of those $k - 1$ letters. Consequently,

$$\Pr(E_k) \geq 1 - \frac{(k - 1)(1 + \varepsilon)}{n}.$$

The choice of a face of i_k is independent of E_k . If i_k chooses its first occurrence and E_k holds, then i_k is first. Writing m_i for the face counts gives

$$m_{i_k} \geq \max \left\{ 0, \frac{n}{1 + \varepsilon} - (k - 1) \right\}. \quad (40)$$

Summing over k yields total length $\Omega(n^2)$ for fixed ε . Condition (36) implies the marginal hypothesis by summing over the orders in which a specified die is first. Theorem 6.1 gives total length $O(n^{23/10}(\log n)^{1/5})$ for fixed positive error, leaving a factor $O(n^{3/10}(\log n)^{1/5})$ between these upper and lower bounds.

7 Conclusion and open problems

Permutation fairness imposes several distinct size scales. The Hahn-polynomial construction gives $2^{O(n \log n)}$ faces on each die, and the arithmetic lower bound forces $2^{\Omega(n)}$ faces on most dice. The minimum for one prescribed die is exactly quadratic in order: $g(n) = \Theta(n^2)$, with every sufficiently large quadratic size realizable. For fixed relative error, independently relabeled palindrome blocks give $O(n^{13/10}(\log n)^{1/5})$ faces on every die.

The Hahn construction uses Wilson's quadrature weights and estimates of Zaremba and Wilson; its arithmetic ingredient is an $O(n \log n)$ -bit common denominator and reuse of one rule throughout the insertion process. The prescribed-die construction instead corrects a rational approximation to real equal-weight quadrature. These constructions leave the following questions open.

1. What is the asymptotic minimum total number A_n of faces in a permutation-fair family of n dice? The present bounds are $2^{\Omega(n)} \leq A_n \leq 2^{O(n \log n)}$.

2. Let E_n be the minimum total number of faces when all dice have the same size. Is $E_n \leq A_n^C$ for an absolute constant C ? For permutation-fair families, equal composition is equivalent to $(\leq n, n)$ -fairness by Lemma 2.9. The existing absolute bounds already imply

$$A_n \leq E_n \leq A_n^{O(\log n)} = A_n^{O(\log \log A_n)},$$

using $\log A_n = \Omega(n)$. A polynomial relation remains open.

3. Can two prescribed dice simultaneously have $O(n^2)$ faces? More generally, how many prescribed dice can be kept at the optimal quadratic scale? Theorem 4.3 controls only one die.
4. For a fixed $0 < \varepsilon < 1$, let $M_n(\varepsilon)$ be the minimum common number of faces for pointwise relative error at most ε . Section 6 gives

$$\Omega_\varepsilon(n) \leq M_n(\varepsilon) \leq O_\varepsilon\left(n^{13/10}(\log n)^{1/5}\right).$$

Can the upper bound be reduced to $O_\varepsilon(n \text{ polylog } n)$, or even $O_\varepsilon(n)$?

Acknowledgments. We are especially grateful to Austin Conner, who devised the first version of our second construction (Section 3) using polynomial interpolation. We thank Florian Haeupler for suggesting the problem, and James Grime, Eric Harshbarger, Michael Purcell, Josef Tkadlec, and commenters on the Polylog video [71] for helpful discussions. T.G. acknowledges the support of Charles University grant PRIMUS/22/HUM/020. V.R. was supported by the Czech Science Foundation (GAČR), JUNIOR STAR grant no. 26-23599M.

AI use. We collaborated with OpenAI’s GPT models and Anthropic’s Claude on mathematical exploration, proof development and checking, and manuscript drafting and editing. The authors take responsibility for the mathematical claims and the final text.

A The moment estimate for random palindrome blocks

We prove Lemma 6.2. Throughout this appendix, $n \geq 3$, $2 \leq p \leq n^2$, and $m \geq 8n^{11/10}p^{1/5}$. All norms and expectations over blocks refer to the independent uniform permutations in (38), unless another random variable is indicated explicitly.

A.1 Two auxiliary estimates

Lemma A.1 (Bounded multilinear chaos). *Let (Z_v) be independent centered random variables with $|Z_v| \leq b_v$. If*

$$T = \sum_{|A|=\ell} c_A \prod_{v \in A} Z_v,$$

then, for $p \geq 2$,

$$\|T\|_p \leq (2\sqrt{p-1})^\ell \left(\sum_{|A|=\ell} c_A^2 \prod_{v \in A} b_v^2 \right)^{1/2}.$$

Proof. Introduce independent copies Z'_v and replace every Z_v in T by $Z_v - Z'_v$. Conditional expectation over the copies recovers T , so Jensen's inequality bounds the original norm by the new one. Swapping each pair (Z_v, Z'_v) independently preserves their joint law and changes the corresponding difference by a Rademacher sign. Conditional on the pairs, the degree- ℓ Rademacher hypercontractive inequality [51, 52] bounds the L^p norm over these signs by $(p-1)^{\ell/2}$ times the coefficient ℓ_2 norm. Since $|Z_v - Z'_v| \leq 2b_v$, this gives the displayed bound uniformly in the pairs, and hence also after averaging over them. $\square$

For a probability vector $u = (u_0, \dots, u_\ell)$, let

$$C_d(u) = \sum_{a_0 + \dots + a_\ell = d} \left(\frac{d!}{a_0! \dots a_\ell!} \prod_{j=0}^{\ell} u_j^{a_j} \right)^2$$

be the collision probability of the multinomial distribution with d trials and cell probabilities u . We use the convention $0^0 = 1$.

Lemma A.2 (A lattice collision bound). *For integers $\ell \geq 1$, $d \geq 0$, and $M \geq d+1$,*

$$\sum_{g_0 + \dots + g_\ell = M} C_d(g/M) \leq \left(\frac{108M}{\sqrt{d+1}} \right)^\ell,$$

where the sum is over nonnegative integer vectors g .

Proof. Elementary Stirling bounds give

$$\max_k \Pr(\text{Poisson}(\lambda) = k) \leq \frac{2}{\sqrt{1+\lambda}}, \quad \Pr(\text{Poisson}(d) = d) \geq \frac{1}{3\sqrt{d+1}}.$$

The cases $\lambda \leq 1$ and $d = 0$ are immediate. For $\lambda > 1$, the maximum is attained at $k = \lfloor \lambda \rfloor$, with $k \geq \lambda/2$, and $k! \geq \sqrt{2\pi k}(k/e)^k$ gives the first inequality. For integer $d \geq 1$, $d! \leq e\sqrt{d}(d/e)^d$ gives the second.

Condition independent Poisson variables of means du_j on their sum being d . Since a collision probability is at most the maximum probability mass, the preceding bounds give

$$C_d(u) \leq 3 \cdot 2^{\ell+1} \sqrt{d+1} \prod_{j=0}^{\ell} (1 + du_j)^{-1/2}.$$

For every g in the sum, some coordinate satisfies $g_j \geq M/(\ell + 1)$. For that coordinate,

$$\sqrt{d+1}(1 + dg_j/M)^{-1/2} \leq \sqrt{\ell+1}.$$

Sum over the $\ell + 1$ possible choices of this coordinate, then drop the sum constraint on the remaining ℓ coordinates. Each of them lies between 0 and M . Since $w(x) = (1 + dx/M)^{-1/2}$ is decreasing,

$$\sum_{g=0}^M w(g) \leq 1 + \int_0^M w(x) dx \leq 1 + \frac{2M}{\sqrt{d+1}} \leq \frac{3M}{\sqrt{d+1}}.$$

The integral bound also holds for $d = 0$; the last inequality uses $M \geq d + 1$. Therefore the sum in the lemma is at most

$$6(\ell + 1)^{3/2} \left(\frac{6M}{\sqrt{d+1}} \right)^\ell \leq \left(\frac{108M}{\sqrt{d+1}} \right)^\ell.$$

For the final inequality, use $\ell + 1 \leq 2^\ell$ and $6 \leq 6^\ell$. $\square$

A.2 The expansion and a coloring argument

Fix $\sigma \in S_n$ and write $R_\sigma = n!P_W(\sigma)$. We use the quantities $\delta_i(A)$ in (39). A prescribed order of k distinct letters occurs at most twice as a subsequence of a palindrome $\rho_i \rho_i^{\text{rev}}$. Indeed, in the ordering defined by ρ_i , the prescribed order must increase and then decrease, and the only possible ambiguity is which occurrence of its largest letter is chosen. Hence, for $k \geq 3$,

$$|\delta_i(A)| \leq b_k := k!2^{1-k}. \quad (41)$$

Here $q_i(A) \leq 2^{1-k}$ and $b_k \geq 1$.

Represent ρ_i by the relative order of independent continuous priorities $Y_{i,1}, \dots, Y_{i,n}$, indexed by positions in σ . Write $\delta_{i,a,k}$ for the segment starting at position a and having length k . For fixed k and a fixed residue class of a modulo k , these variables are independent: their segments use disjoint sets of priorities. Different blocks are independent as well.

Expanding (39), the part involving exactly ℓ nonconstant factors has the form

$$T_\ell = \sum_{I, \mathbf{k}, \mathbf{a}} c(I, \mathbf{k}, \mathbf{a}) \prod_{j=1}^{\ell} \delta_{i_j, a_j, k_j}, \quad I = \{i_1 < \dots < i_\ell\}, \quad k_j \geq 3.$$

The coefficients are nonnegative probabilities of the corresponding occupancies and segment starts. The selected segments are disjoint and appear in their natural order. The sum of these homogeneous parts is $R_\sigma - 1$.

For any fixed restriction $\mathcal{D}$ on these indices, we claim that

$$\|T_\ell^\mathcal{D}\|_p \leq (2\sqrt{p-1})^\ell \left[\sum_{(I, \mathbf{k}, \mathbf{a}) \in \mathcal{D}} c(I, \mathbf{k}, \mathbf{a})^2 \prod_{j=1}^{\ell} \frac{k_j b_{k_j}^2}{q_{k_j}} \right]^{1/2}, \quad q_k = 2^{-(k-2)} \quad (k \geq 3). \quad (42)$$

Independently for each block, choose a size K_i with probabilities q_k , and a uniform residue B_i modulo K_i . Keep only terms with $k_j = K_{i_j}$ and $a_j \equiv B_{i_j} \pmod{k_j}$, multiplying their coefficients by $\prod_j k_j/q_{k_j}$. Averaging over these auxiliary choices recovers the original sum. Given the choices, all remaining variables are independent and centered, so Lemma A.1 applies. Minkowski's inequality followed by Jensen's inequality for the square root proves (42): a term is retained with probability $\prod_j q_{k_j}/k_j$. A choice $K_i > n$ simply retains no terms from block i .

We next compute the coefficients. For fixed I and $\mathbf{k}$, put $s = \sum_j k_j$, $d = n - s$, and

$$g_0 = i_1 - 1, \quad g_j = i_{j+1} - i_j - 1 \quad (1 \leq j < \ell), \quad g_\ell = m - i_\ell, \quad M = m - \ell.$$

The unspecified blocks lie in these $\ell + 1$ gaps and $\sum_j g_j = M$. The segment starts are equivalently specified by the numbers $a_0, \dots, a_\ell$ of remaining letters assigned to those gaps, where $\sum_j a_j = d$. Direct summation of the multinomial law gives

$$c(I, \mathbf{k}, \mathbf{a}) = \frac{n^s}{m^s \prod_j k_j!} \left(\frac{M}{m}\right)^d \frac{d!}{a_0! \cdots a_\ell!} \prod_{j=0}^{\ell} \left(\frac{g_j}{M}\right)^{a_j}. \quad (43)$$

Here and below $\mathbf{a}$ denotes gap counts rather than the equivalent segment starts. The map $I \mapsto \mathbf{g}$ is a bijection between ℓ -subsets of $[m]$ and weak compositions of $m - \ell$ into $\ell + 1$ parts. Since $m \geq n$ and $\ell \leq n/3$, we always have $M > 0$.

A.3 Terms with short total support

Split the expansion into terms with $s \leq n/2$ and $s > n/2$, denoting their sums by T^{short} and T^{long} . In the first case, $d \geq n/2$ and $M = m - \ell \geq d + 2\ell \geq d + 1$. Using (43), (41), and $n^s \leq n^s$, the square sum in (42) for this part is at most

$$\sum_{\mathbf{k}: s \leq n/2} \prod_{j=1}^{\ell} \left[\left(\frac{n}{m}\right)^{2k_j} 2^{2-2k_j} \frac{k_j}{q_{k_j}} \right] \sum_{g_0 + \cdots + g_\ell = M} C_d(g/M).$$

We have also used $(M/m)^{2d} \leq 1$. Lemma A.2 bounds the collision sum by $(160m/\sqrt{n})^\ell$, because $108\sqrt{2} < 160$. Extend the length sum to all $k_j \geq 3$ to separate it into a product. For $z = n^2/(2m^2) \leq 1/2$,

$$\sum_{k \geq 3} \left(\frac{n}{m}\right)^{2k} 2^{2-2k} \frac{k}{q_k} = \sum_{k \geq 3} k z^k = \frac{z^3(3-2z)}{(1-z)^2} \leq 8z^3 = \frac{n^6}{m^6}.$$

Consequently, if

$$a = 2\sqrt{160} \frac{n^{11/4} \sqrt{p}}{m^{5/2}},$$

then

$$\|T^{\text{short}}\|_p \leq \sum_{\ell \geq 1} a^\ell = \frac{a}{1-a} \quad (a < 1). \quad (44)$$

A.4 Terms with long total support

For $s > n/2$, bound each collision probability by one and sum over I , using $\binom{m}{\ell} \leq m^\ell/\ell!$. For any $\theta \geq 1$, the factor $\theta^{s-n/2}$ is at least one. Inserting this factor into the nonnegative square sum in (42) gives

$$\|T_\ell^{\text{long}}\|_p \leq \theta^{-n/4} \frac{A_\theta^\ell}{\sqrt{\ell!}}, \quad A_\theta = 2\sqrt{pm} \left(\sum_{k \geq 3} k \left(\frac{\theta n^2}{2m^2} \right)^k \right)^{1/2}.$$

Set $t = m/n \geq 1$ and $\theta = t^{1/3}$. The power-series argument is $1/(2t^{5/3}) \leq 1/2$, and hence

$$A_\theta^2 \leq 4pm t^{-5} = \frac{4pn}{t^4}.$$

Cauchy–Schwarz gives

$$\sum_{\ell \geq 0} \frac{A^\ell}{\sqrt{\ell!}} \leq \left(\sum_{\ell \geq 0} 2^{-\ell} \right)^{1/2} \left(\sum_{\ell \geq 0} \frac{2^\ell A^{2\ell}}{\ell!} \right)^{1/2} = \sqrt{2} e^{A^2}.$$

Therefore

$$\|T^{\text{long}}\|_p \leq \sqrt{2} \exp \left(-\frac{n}{12} \log t + \frac{4pn}{t^4} \right). \quad (45)$$

A.5 Completion of the moment bound

Put

$$H = \frac{n^{11/4} \sqrt{p}}{m^{5/2}}.$$

The assumption on m gives $H \leq 8^{-5/2}$, $a \leq 1/6$, and $t \geq 8n^{1/10}p^{1/5} \geq 8$. Thus (44) yields

$$\|T^{\text{short}}\|_p \leq \frac{6}{5}a = \frac{12}{5}\sqrt{160} H.$$

Moreover, since $p \leq n^2$,

$$\frac{4p}{t^4} \leq \frac{1}{1024} \left(\frac{p}{n^2} \right)^{1/5} \leq \frac{1}{1024} \leq \frac{\log t}{24}.$$

For $n \geq 60$, (45) consequently gives

$$\|T^{\text{long}}\|_p \leq \sqrt{2} t^{-n/24} \leq \sqrt{2} t^{-5/2} \leq H,$$

where the last inequality uses $H = n^{1/4} \sqrt{p} t^{-5/2}$. Combining the two parts gives $\|R_\sigma - 1\|_p \leq 32H$ for $n \geq 60$.

For completeness, the remaining $3 \leq n < 60$ are covered uniformly by a simpler estimate that keeps each entire block as one variable. Let F_I be the part of the product expansion using exactly the nonconstant factors indexed by I . This is a function of those blocks, centered separately in each block. The selected-cell multinomial marginal and (41) give

$$\|F_I\|_\infty \leq \beta^{|I|}, \quad \beta = \sum_{k \geq 3} 2^{1-k} (n/m)^k = \frac{n^3}{4m^3(1 - n/(2m))} \leq \frac{n^3}{2m^3}.$$

Indeed, the joint probability of selected occupancies k_i with sum s is bounded by

$$\frac{n^s}{m^s \prod_i k_i!}.$$

The factorials cancel against (41), and extending the sum to arbitrary $k_i \geq 3$ gives the product above.

To bound the sum over $|I| = \ell$, introduce an independent copy of each block. Replace F_I by its alternating difference in its ℓ arguments, denoted by $\Delta_I F_I$. Separate centering implies that averaging over the copies recovers F_I , while $|\Delta_I F_I| \leq 2^\ell \beta^\ell$. Independently swapping each block with its copy multiplies $\Delta_I F_I$ by the product of the corresponding Rademacher signs. Conditional hypercontractivity, followed by the uniform bound on these coefficients, gives

$$\left\| \sum_{|I|=\ell} F_I \right\|_p \leq (2\sqrt{p-1})^\ell \binom{m}{\ell}^{1/2} \beta^\ell \leq \frac{b^\ell}{\sqrt{\ell!}}, \quad b = 2\beta\sqrt{mp}.$$

Since $b \leq n^{1/4}H \leq 60^{1/4}8^{-5/2} < 1/4$, we obtain

$$\|R_\sigma - 1\|_p \leq \sum_{\ell \geq 1} \frac{b^\ell}{\sqrt{\ell!}} \leq \frac{b}{1-b} \leq \frac{4}{3} 60^{1/4}H < 4H.$$

This proves Lemma 6.2 for all $n \geq 3$.

References

- [1] Eric Harshbarger. “Go first” dice, n.d. URL https://www.ericharshbarger.org/dice/go_first_dice.html.
- [2] Eric Harshbarger. Who goes first. *Games Magazine*, 36(10), December 2012.
- [3] Robert Ford, James Grime, Eric Harshbarger, and Brian Pollock. Go first dice for five players and beyond. *Recreational Mathematics Magazine*, 10(17):75–87, 2023. doi: 10.2478/rmm-2023-0004.
- [4] Michael Purcell. Building sets of permutation-fair dice. *American Mathematical Monthly*, 131(7):595–607, 2024. doi: 10.1080/00029890.2024.2345576.
- [5] Byron Knoll. Five player go first dice. Blog post, 2012. URL <http://byronknoll.blogspot.se/2012/10/five-player-go-first-dice.html>.
- [6] The Dice Lab. 5-player go first dice (construction discovered by Paul Meyer), n.d. URL <https://www.mathartfun.com/thedicelab.com/GFD5.html>. Accessed July 20, 2026.
- [7] Vašek Chvátal, David A. Klarner, and Donald E. Knuth. Selected combinatorial research problems. Technical report, Computer Science Department, Stanford University, 1972.
- [8] Malcolm Newey. Notes on a problem involving permutations as subsequences. Technical report, Computer Science Department, Stanford University, 1973.
- [9] Leonard Adleman. Short permutation strings. *Discrete Mathematics*, 10(2):197–200, 1974.
- [10] P. J. Koutas and T. C. Hu. Shortest string containing all permutations. *Discrete Mathematics*, 11(2):125–132, 1975.
- [11] Giulia Galbiati and Franco P. Preparata. On permutation-embedding sequences. *SIAM Journal on Applied Mathematics*, 30(3):421–423, 1976.
- [12] Daniel J. Kleitman and David J. Kwiatkowski. A lower bound on the length of a sequence containing all permutations as subsequences. *Journal of Combinatorial Theory, Series A*, 21(2):129–136, 1976.
- [13] Eugen Zălinescu. Shorter strings containing all k -element permutations. *Information Processing Letters*, 111(12):605–608, 2011.
- [14] Saša Radomirović. A construction of short sequences containing all permutations of a set as subsequences. *Electronic Journal of Combinatorics*, 19(4):P31, 2012.

- [15] Michael Engen and Vincent Vatter. Containing all permutations. *American Mathematical Monthly*, 128(1):4–24, 2021. doi: 10.1080/00029890.2021.1835384.
- [16] Gal Beniamini, Nir Lavee, and Nati Linial. How balanced can permutations be? *Combinatorica*, 45:9, 2025. doi: 10.1007/s00493-024-00127-x.
- [17] Anton Černý. On fairness of D0L systems. *Discrete Applied Mathematics*, 155(13):1769–1773, 2007. doi: 10.1016/j.dam.2007.04.004.
- [18] Anton Černý. On fair words. *Journal of Automata, Languages and Combinatorics*, 14(2):163–174, 2009. doi: 10.25596/JALC-2009-163.
- [19] Helmut Prodinger. On a generalization of the Dyck-language over a two letter alphabet. *Discrete Mathematics*, 28(3):269–276, 1979. doi: 10.1016/0012-365X(79)90134-1.
- [20] Anton Černý. On subword symmetry of words. *International Journal of Foundations of Computer Science*, 19(1):243–250, 2008. doi: 10.1142/S0129054108005656.
- [21] Arto Salomaa. Subword balance in binary words, languages and sequences. *Fundamenta Informaticae*, 75(1–4):469–482, 2007. doi: 10.3233/FUN-2007-751-427.
- [22] Gwenaël Richomme. On some 2-binomial coefficients of binary words: geometrical interpretation, partitions of integers, and fair words. *Theoretical Computer Science*, 1066:115732, 2026. doi: 10.1016/j.tcs.2025.115732.
- [23] Brian G. Kronenthal and Lorenzo Traldi. Tied dice. *Journal of Combinatorial Mathematics and Combinatorial Computing*, 68:85–96, 2009. URL <https://combinatorialpress.com/jcmcc-articles/volume-068/tied-dice/>.
- [24] C. Cooley, W. Ella, M. Follett, E. Gilson, and L. Traldi. Tied dice II: some asymptotic results. *Journal of Combinatorial Mathematics and Combinatorial Computing*, 90:241–248, 2014. URL <https://combinatorialpress.com/jcmcc-articles/volume-090/tied-dice-ii-some-asymptotic-results/>.
- [25] Alex Schaefer and Jay Schweig. Balanced nontransitive dice. *College Mathematics Journal*, 48(1):10–16, 2017. doi: 10.4169/college.math.j.48.1.10.
- [26] Injo Hur and Yeansu Kim. Possible probability and irreducibility of balanced nontransitive dice. *Journal of Mathematics*, 2021:6648248, 2021. doi: 10.1155/2021/6648248.
- [27] Dohyeon Kim, Ringi Kim, Wonjun Lee, Yuhyeon Lim, and Yoojin So. Balanced nontransitive dice: existence and probability. *Electronic Journal of Combinatorics*, 31(1):P1.21, 2024. doi: 10.37236/11918.
- [28] M. Wayne Wilson. A general algorithm for nonnegative quadrature formulas. *Mathematics of Computation*, 23:253–258, 1969. doi: 10.1090/S0025-5718-1969-0242374-1.
- [29] M. Wayne Wilson. Discrete least squares and quadrature formulas. *Mathematics of Computation*, 24(110):271–282, 1970. doi: 10.1090/S0025-5718-1970-0275677-3.
- [30] M. Wayne Wilson. Necessary and sufficient conditions for equidistant quadrature formula. *SIAM Journal on Numerical Analysis*, 7(1):134–141, 1970. doi: 10.1137/0707009.
- [31] M. Wayne Wilson. On the Hahn polynomials. *SIAM Journal on Mathematical Analysis*, 1(1):131–139, 1970. doi: 10.1137/0501013.

- [32] Daan Huybrechs. Stable high-order quadrature rules with equidistant points. *Journal of Computational and Applied Mathematics*, 231(2):933–947, 2009. doi: 10.1016/j.cam.2009.05.018.
- [33] Jan Glaubitz. Stable high order quadrature rules for scattered data and general weight functions. *SIAM Journal on Numerical Analysis*, 58:2144–2164, 2020. doi: 10.1137/19M1257901.
- [34] Francesca Cappelazzo, Wolfgang Erb, Alfredo E. Marchetti, and Davide Poggiali. On Kosloff Tal-Ezer least-squares quadrature formulas. *BIT Numerical Mathematics*, 63: 15, 2023. doi: 10.1007/s10543-023-00948-0.
- [35] L. Fejér. Mechanische quadraturen mit positiven cotesschen zahlen. *Mathematische Zeitschrift*, 37:287–309, 1933.
- [36] J. Shohat. On mechanical quadratures, in particular, with positive coefficients. *Transactions of the American Mathematical Society*, 42:461–496, 1937.
- [37] S. K. Zaremba. Some properties of polynomials orthogonal over the set $\langle 1, 2, \dots, N \rangle$. *Annali di Matematica Pura ed Applicata, Series IV*, 105:333–345, 1975. doi: 10.1007/BF02414937.
- [38] Holger Dette. New bounds for Hahn and Krawtchouk polynomials. *SIAM Journal on Mathematical Analysis*, 26(6):1647–1659, 1995. doi: 10.1137/S0036141093255661.
- [39] P. D. Seymour and T. Zaslavsky. Averaging sets: a generalization of mean values and spherical designs. *Advances in Mathematics*, 52(3):213–240, 1984. doi: 10.1016/0001-8708(84)90022-7.
- [40] Daniel M. Kane. Small designs for path-connected spaces and path-connected homogeneous spaces. *Transactions of the American Mathematical Society*, 367:6387–6414, 2015. doi: 10.1090/tran/6250.
- [41] S. N. Bernstein. Sur les formules de quadrature de Cotes et Tchebycheff; and on quadrature formulas with positive coefficients. *C. R. Acad. Sci. URSS*, 14:323–327; and *Izv. Akad. Nauk SSSR Ser. Mat.*, 1(4):479–503, 1937.
- [42] A. B. J. Kuijlaars. The minimal number of nodes in Chebyshev type quadrature formulas. *Indagationes Mathematicae*, 4(3):339–362, 1993. doi: 10.1016/0019-3577(93)90007-L.
- [43] A. B. J. Kuijlaars. Chebyshev-type quadrature for Jacobi weight functions. *Journal of Computational and Applied Mathematics*, 57:171–180, 1995. doi: 10.1016/0377-0427(93)E0243-F.
- [44] Shoni Gilboa and Ron Peled. Chebyshev-type quadratures for doubling weights. *Constructive Approximation*, 45(2):193–216, 2017. doi: 10.1007/s00365-016-9360-4.
- [45] Ron Peled. Simple universal bounds for Chebyshev-type quadratures. *Journal of Approximation Theory*, 162(12):2317–2348, 2010. doi: 10.1016/j.jat.2010.08.002.
- [46] Zhen Cui, Jiacheng Xia, and Ziqing Xiang. Rational designs. *Advances in Mathematics*, 352:541–571, 2019. doi: 10.1016/j.aim.2019.06.012.

- [47] G. I. Arkhipov. On the Hilbert–Kamke problem. *Mathematics of the USSR-Izvestiya*, 24(1):1–47, 1985. doi: 10.1070/IM1985v024n01ABEH001210.
- [48] Richard P. Stanley. *Ordered Structures and Partitions*. Number 119 in Memoirs of the American Mathematical Society. American Mathematical Society, 1972. doi: 10.1090/memo/0119.
- [49] Dave Bayer and Persi Diaconis. Trailing the dovetail shuffle to its lair. *Annals of Applied Probability*, 2(2):294–313, 1992. doi: 10.1214/aoap/1177005705.
- [50] Persi Diaconis, Jason Fulman, and Susan Holmes. Analysis of casino shelf shuffling machines. *Annals of Applied Probability*, 23(4):1692–1720, 2013. doi: 10.1214/12-AAP884.
- [51] Aline Bonami. Étude des coefficients de Fourier des fonctions de $L^p(G)$. *Annales de l’Institut Fourier*, 20(2):335–402, 1970. doi: 10.5802/aif.357.
- [52] Ryan O’Donnell. *Analysis of Boolean Functions*. Cambridge University Press, 2014. doi: 10.1017/CBO9781139814782.
- [53] G. G. Lorentz. *Bernstein Polynomials*. Chelsea Publishing Company, 1986.
- [54] S. Karlin and J. McGregor. The Hahn polynomials, formulas and an application. *Scripta Mathematica*, 26:33–46, 1961.
- [55] Roelof Koekoek, Peter A. Lesky, and René F. Swarttouw. *Hypergeometric Orthogonal Polynomials and Their q -Analogues*. Springer Monographs in Mathematics. Springer, 2010. doi: 10.1007/978-3-642-05014-5.
- [56] NIST Digital Library of Mathematical Functions. Chapter 18: Orthogonal polynomials, sections 18.19–18.20, n.d. URL <https://dlmf.nist.gov/18.19>. Accessed July 2026.
- [57] Steven R. Finch. *Mathematical Constants*. Cambridge University Press, 2003.
- [58] Eugène Prouhet. Mémoire sur quelques relations entre les puissances des nombres. *Comptes rendus de l’Académie des sciences de Paris*, 33:225–228, 1851.
- [59] Ajai Choudhry. An improvement of Prouhet’s 1851 result on multigrade chains. *International Journal of Number Theory*, 16(7):1425–1432, 2020. doi: 10.1142/S179304212050075X.
- [60] D. H. Lehmer. The Tarry–Escott problem. *Scripta Mathematica*, 13:37–41, 1947.
- [61] E. M. Wright. Equal sums of like powers. *Proceedings of the Edinburgh Mathematical Society, Series 2*, 8:138–142, 1949.
- [62] E. M. Wright. Prouhet’s 1851 solution of the Tarry–Escott problem of 1910. *American Mathematical Monthly*, 66(3):199–201, 1959. doi: 10.1080/00029890.1959.11989269.
- [63] J. B. Roberts. Splitting consecutive integers into classes with equal power sums. *American Mathematical Monthly*, 71:25–37, 1964. doi: 10.1080/00029890.1964.11992196.
- [64] J. B. Roberts. Relations between the digits of numbers and equal sums of like powers. *Canadian Journal of Mathematics*, 16:626–636, 1964. doi: 10.4153/CJM-1964-063-7.

- [65] Ethan Bolker, Carl Offner, Robert Richman, and Catalin Zara. The Prouhet–Tarry–Escott problem and generalized Thue–Morse sequences. *Journal of Combinatorics*, 7(1):117–133, 2016. doi: 10.4310/JOC.2016.v7.n1.a5.
- [66] David W. Boyd. On a problem of Byrnes concerning polynomials with restricted coefficients. *Mathematics of Computation*, 66(220):1697–1703, 1997. doi: 10.1090/S0025-5718-97-00892-2.
- [67] David W. Boyd. On a problem of Byrnes concerning polynomials with restricted coefficients. II. *Mathematics of Computation*, 71(239):1205–1217, 2002. doi: 10.1090/S0025-5718-01-01360-6.
- [68] Daniel Berend and Shahar Golan. Littlewood polynomials with high order zeros. *Mathematics of Computation*, 75(255):1541–1552, 2006. doi: 10.1090/S0025-5718-06-01848-5.
- [69] Joe Buhler, Shahar Golan, Rob Pratt, and Stan Wagon. Littlewood polynomials, spectral-null codes, and equipowerful partitions. *Mathematics of Computation*, 90(329):1435–1453, 2021. doi: 10.1090/mcom/3612.
- [70] Paul Pollack and Enrique Treviño. Partitioning powers into sets of equal sum. *Rocky Mountain Journal of Mathematics*, 56(2):503–514, 2026. doi: 10.1216/rmj.2026.56.503.
- [71] Polylog. We designed special dice using math, but there’s a catch, n.d. URL <https://www.youtube.com/watch?v=-64UT8yikng>.